

9. 光的干涉和干涉系统 (第12章)

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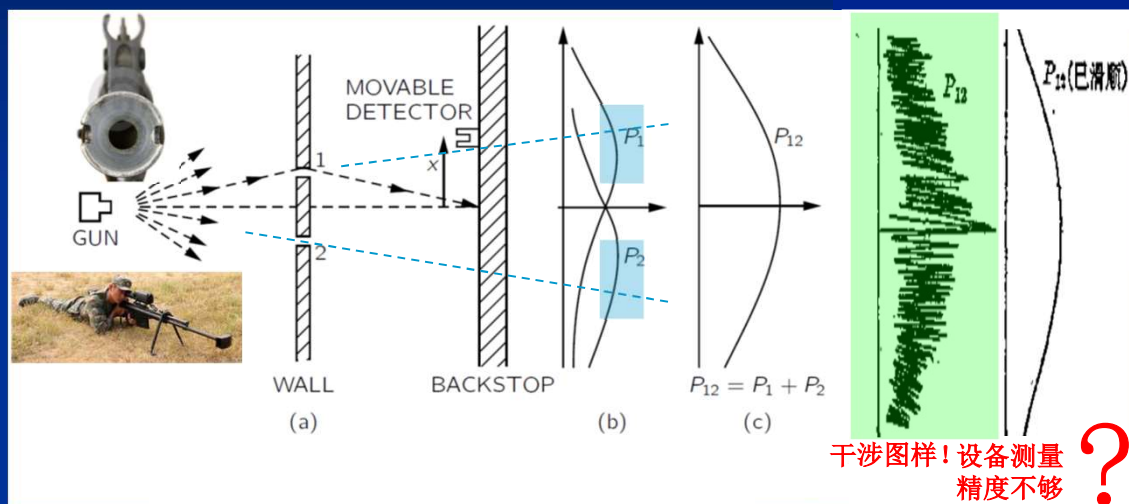
9.7 光子波干涉

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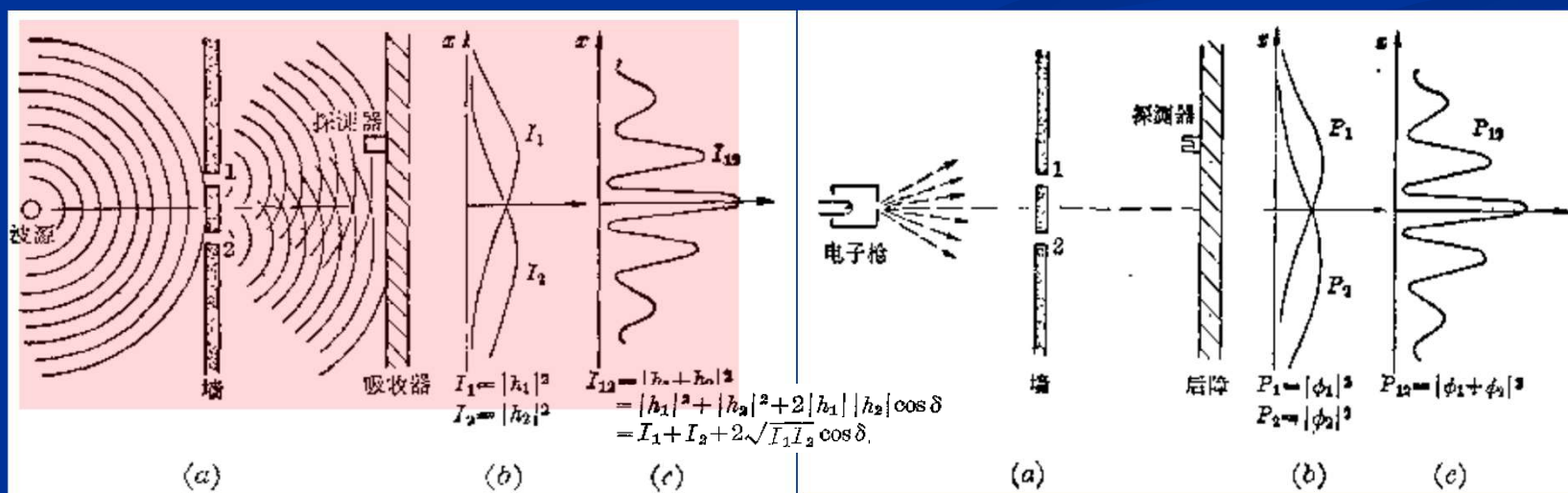
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序言：经典现象

稀疏枪弹射击

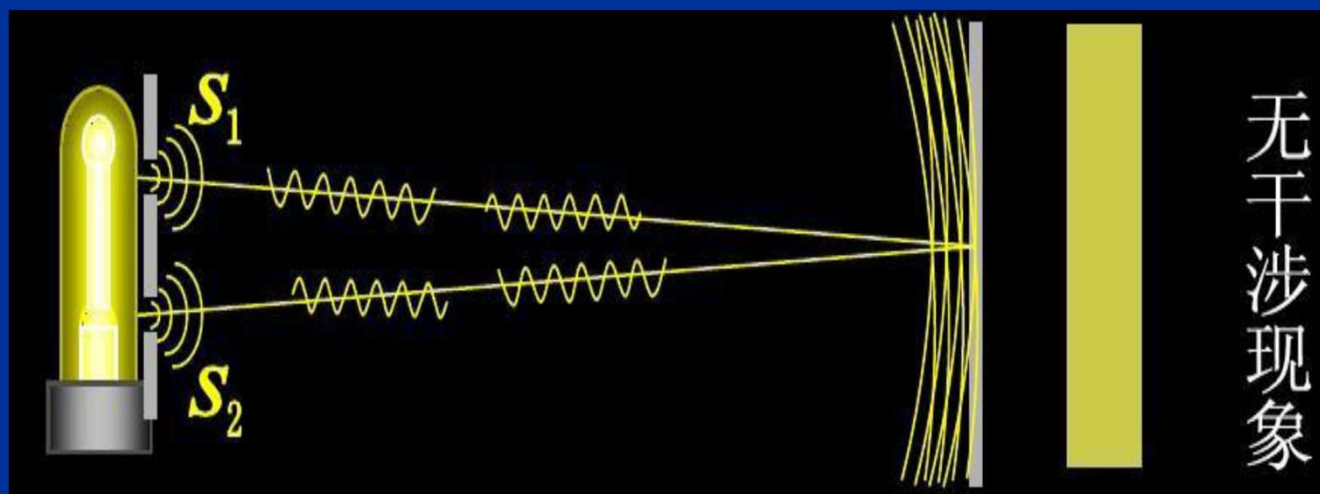
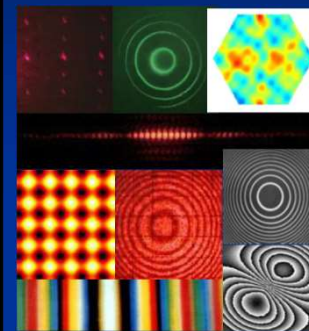
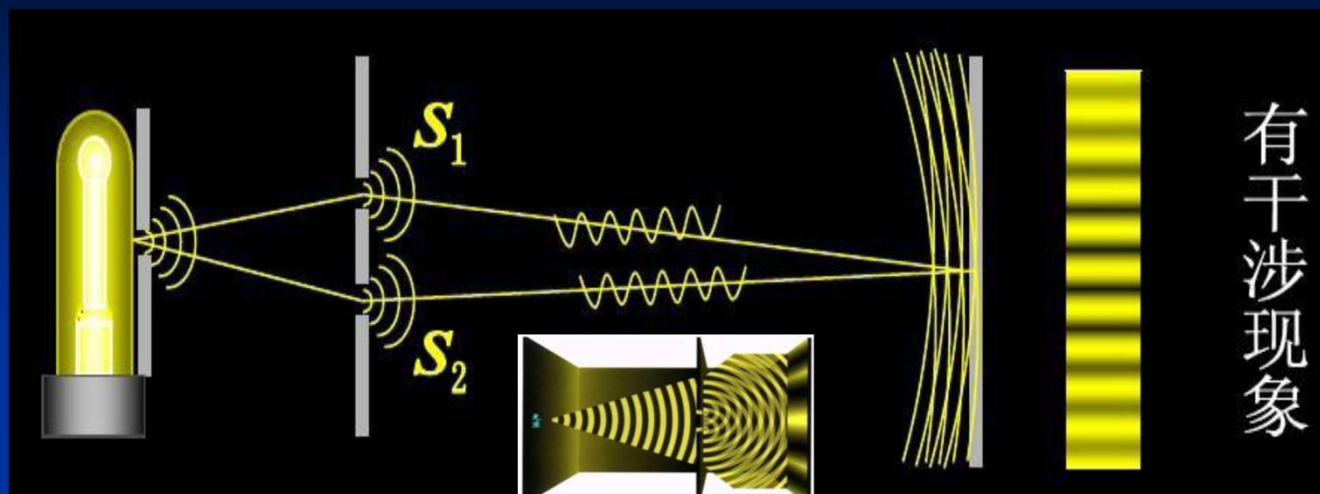


子弹波干涉



水波干涉

电子波干涉



光波干涉：时间累积干涉（常规情形）
瞬态干涉

问题是什么？

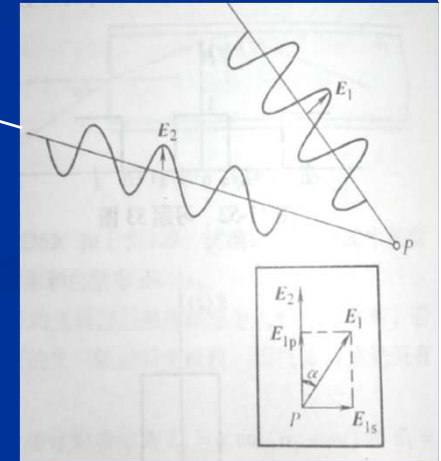
- 反映：能流的→时间累积
→空间再分布（积聚、弱化）→空间调制
 - 唯像表征？
 - 内在原因？
- 使用条件：光程差稳定（长时、瞬态）
 - 外差：频率稳定
 - 同频：光程差稳定
 - 探测器帧频足够高：任何光波、任何频率

9.1 光波干涉的条件

$$\langle f(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(t) dt$$

$$E_1(r, t) = A_1 \exp \left[j(\vec{k}_1 \cdot \vec{r}) - \omega_1 t + \delta_1 \right]$$

$$E_2(r, t) = A_2 \exp \left[j(\vec{k}_2 \cdot \vec{r}) - \omega_2 t + \delta_2 \right]$$



$$E(r, t) = E_1(r, t) + E_2(r, t)$$

$$I = \langle E \cdot E^* \rangle = \left\langle \left[E_1(r, t) + E_2(r, t) \right] \cdot \left[E_1^*(r, t) + E_2^*(r, t) \right] \right\rangle$$

$$= E_1 \cdot E_1^* + E_2 \cdot E_2^* + \left\langle \text{Re} \left[2E_1 \cdot E_2^* \right] \right\rangle$$

$$= A_1^2 + A_2^2 + 2A_1A_2 \cos \alpha \cdot \left\langle \cos \left[(\vec{k}_1 - \vec{k}_2) \cdot \vec{r} + (\delta_1 - \delta_2) - (\omega_1 - \omega_2)t \right] \right\rangle$$

$$= I_1 + I_2 + I_{12}$$

相干项：

$$\begin{aligned} I_{12} &= 2A_1A_2 \cos \alpha \cdot \left\langle \cos \left[\left(\vec{k}_1 - \vec{k}_2 \right) \cdot \vec{r} + \left(\delta_1 - \delta_2 \right) - \left(\omega_1 - \omega_2 \right) t \right] \right\rangle \\ &= 2A_1A_2 \cos \alpha \cdot \langle \cos \psi \rangle \\ &= 2\sqrt{I_1}\sqrt{I_2} \cos \alpha \cdot \langle \cos \psi \rangle \\ &= 2\sqrt{I_1}\sqrt{I_2} \cos \alpha \cdot \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \cos \psi \, dt \end{aligned}$$

$$\psi = \left(\vec{k}_1 - \vec{k}_2 \right) \cdot \vec{r} + \left(\delta_1 - \delta_2 \right) - \left(\omega_1 - \omega_2 \right) t \stackrel{\text{令}}{=} \Delta \vec{k}(t) \cdot \vec{r} + \Delta \delta(t) + \Delta \omega(t) \cdot t = \psi(t)$$

含时情形

$$I_{12} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \psi(t) \, dt = 0 \quad \text{无干涉}$$

典型结果

物质发光: ns级
人眼时间响应常数: 0.1s (亚秒级)
感光胶片: ms级
探测器: ms级、μs级、ns级、ps级

1) 光电探测器: 超快探测器, $(\delta_1 - \delta_2) - (\omega_1 - \omega_2)t$ 为常量, 瞬态干涉

慢探测器: A) 宽频光, $I_{12}=0$, 无干涉

B) $\omega_1 = \omega_2$, $\delta_1 - \delta_2$ 相对稳定 (和时间无关)

$$I_{12} = 2A_1A_2 \cos \alpha \cos \left[(\vec{k}_1 - \vec{k}_2) \cdot \vec{r} + (\delta_1 - \delta_2) \right], \quad \delta_{\text{光程差}} = (\vec{k}_1 - \vec{k}_2) \cdot \vec{r} + (\delta_1 - \delta_2)$$

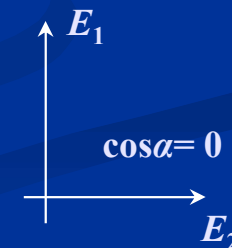
干涉

C) $\omega_1 = \omega_2$, $(\delta_1 - \delta_2)$ 瞬变 (时间的函数)
 $I_{12}=0$, 无干涉

2) 人眼: 慢探测器

典型的光波干涉条件:

- 相同振动方向
- 相同频率
- 相位差恒定
- 同时刻在空间某处相遇



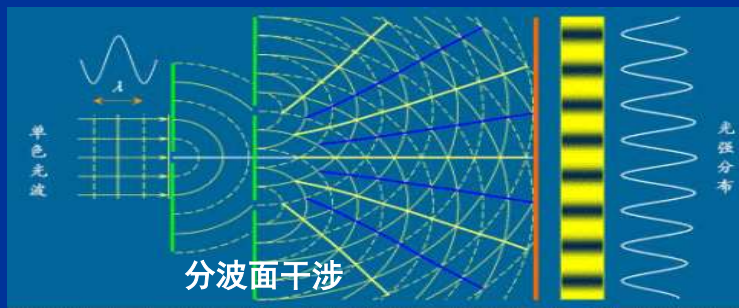
9.2 杨氏 (Thomas Young) 干涉实验

条件: $\omega_1 = \omega_2$, $\delta_1 - \delta_2$ 相对稳定 (和时间无关), $\cos \alpha = 1$

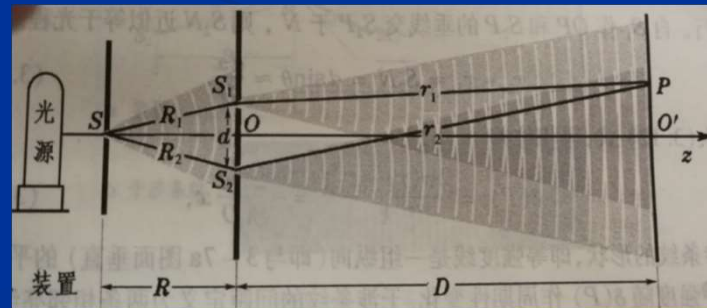
$$I_{12} = 2A_1A_2 \cos \alpha \cdot \cos \left[(\vec{k}_1 - \vec{k}_2) \cdot \vec{r} + (\delta_1 - \delta_2) \right]$$

$$I = I_1 + I_2 + I_{12} = I_1 + I_2 + 2\sqrt{I_1I_2} \cos \left[(\vec{k}_1 - \vec{k}_2) \cdot \vec{r} + (\delta_1 - \delta_2) \right]$$

$$I = 4I_0 \cos^2 \left[\frac{\pi(r_2 - r_1)}{\lambda} \right], \quad (\text{令: } I_1 = I_2 = I_0)$$



$$\delta_1 = \delta_2$$



$$r_1 = S_1 P = \sqrt{\left(x - \frac{d}{2}\right)^2 + y^2 + D^2}$$

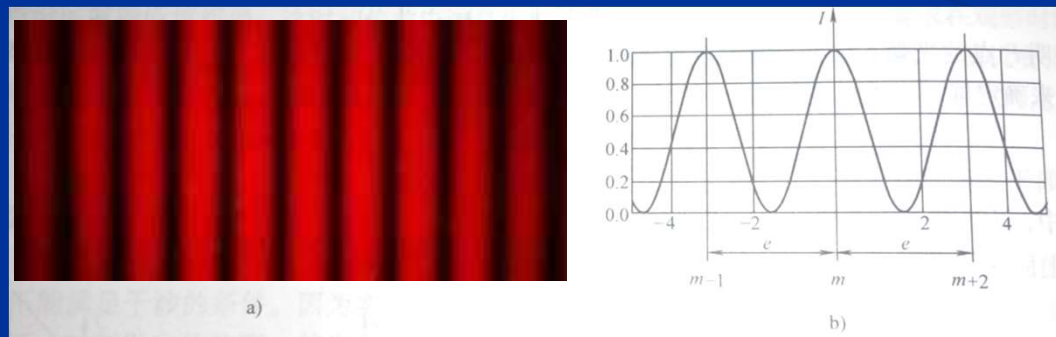
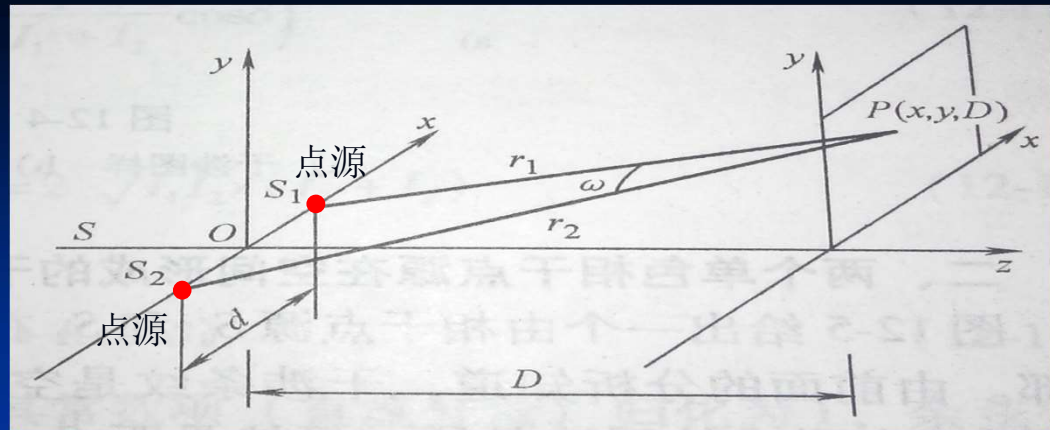
$$r_2 = S_2 P = \sqrt{\left(x + \frac{d}{2}\right)^2 + y^2 + D^2}$$

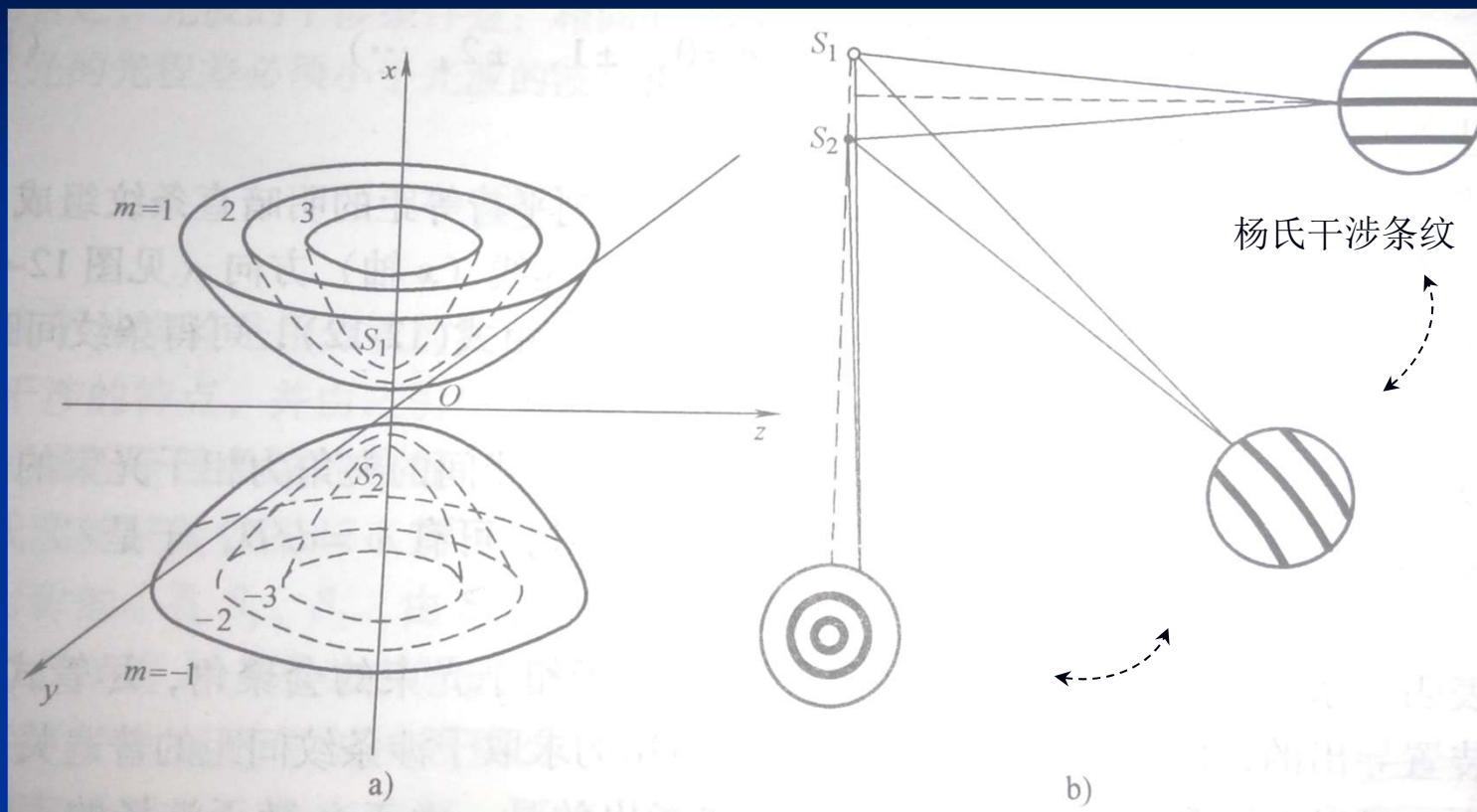
$$\Delta = r_2 - r_1 = \frac{2xd}{r_1 + r_2}$$

若： $d \ll D$ 及 $x, y \ll D$ (远场), $\Delta = \frac{2xd}{r_1 + r_2} \approx \frac{xd}{D}$

$$I = 4I_0 \cos^2 \left[\frac{\pi xd}{\lambda D} \right], \quad x = \frac{mD\lambda}{d} \text{ 时亮纹 (} m=0, \pm 1, \pm 2, \dots \text{)}$$

条纹间距 $e = \frac{D\lambda}{d}$, 相干光束会聚角 $\omega = \frac{d}{D}$





空间干涉场的典型特征

9.3 干涉光场典型特征

可见度定义:

$$K = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$
$$I_{\max} = (A_1 + A_2)^2$$
$$I_{\min} = (A_1 - A_2)^2$$

杨氏干涉可见度:

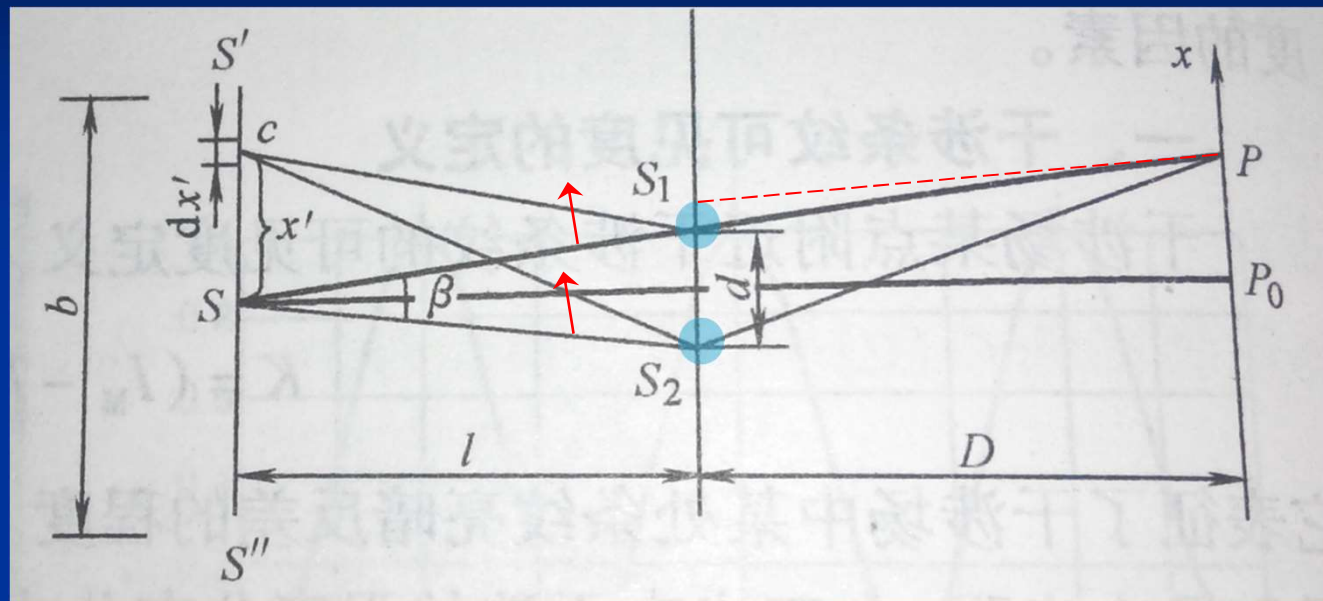
$$K_{\text{杨氏}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} = \frac{2\left(\frac{A_1}{A_2}\right)}{1 + \left(\frac{A_1}{A_2}\right)^2}, \quad \begin{array}{l} \text{若 } A_1 = A_2, K_{\text{杨氏}} = 1 \\ \text{若 } A_1 \neq A_2, K_{\text{杨氏}} < 1 \end{array}$$

$$I = (I_1 + I_2) \left(1 + K_{\text{杨氏}} \cos \delta_{\text{光程差}} \right)$$

$$I = I_{\text{入射}} \left(1 + K_{\text{杨氏}} \cos \delta_{\text{光程差}} \right)$$

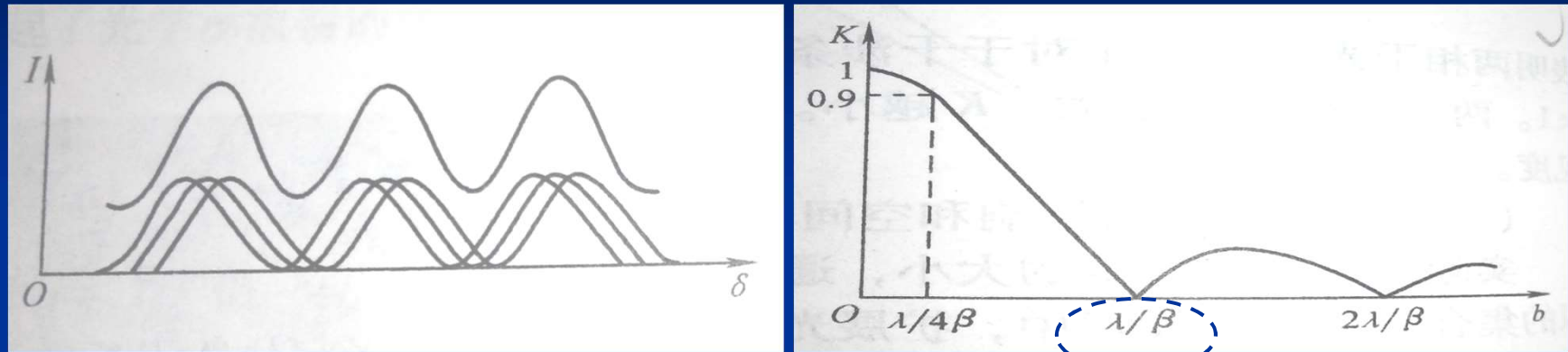
$K_{\text{杨氏}} = 1$, 完全相干,	$I = 2I_{\text{入射}}$
$0 < K_{\text{杨氏}} < 1$, 部分相干,	$I < 2I_{\text{入射}}$
$K_{\text{杨氏}} = 0$, 完全不相干,	$I = I_{\text{入射}}$

1) 面/扩展/源：条纹可见度随光源大小的变化



杨氏干涉可见度： $K_{\text{杨氏}} = \left| \frac{\lambda}{\pi b \beta} \sin \frac{\pi b \beta}{\lambda} \right|$

干涉孔径角： $\beta = \frac{d}{l}$



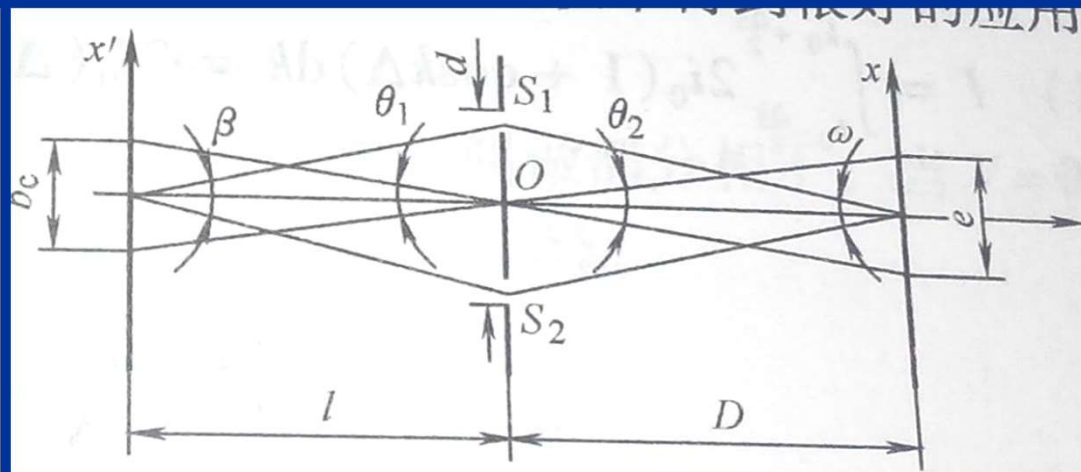
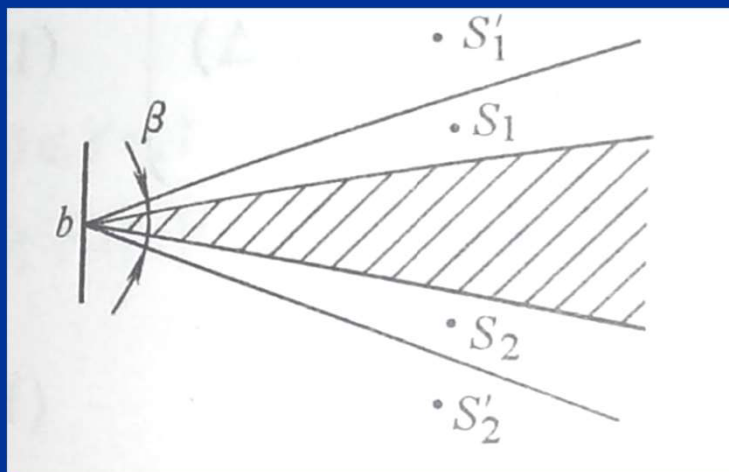
光源临界宽度: $b_c = \frac{\lambda}{\beta}$

光源允许宽度: $b_p = \frac{b_c}{4}$

2) 空间相干性

不变量: $b_c \beta = e \omega = d \theta = \lambda$

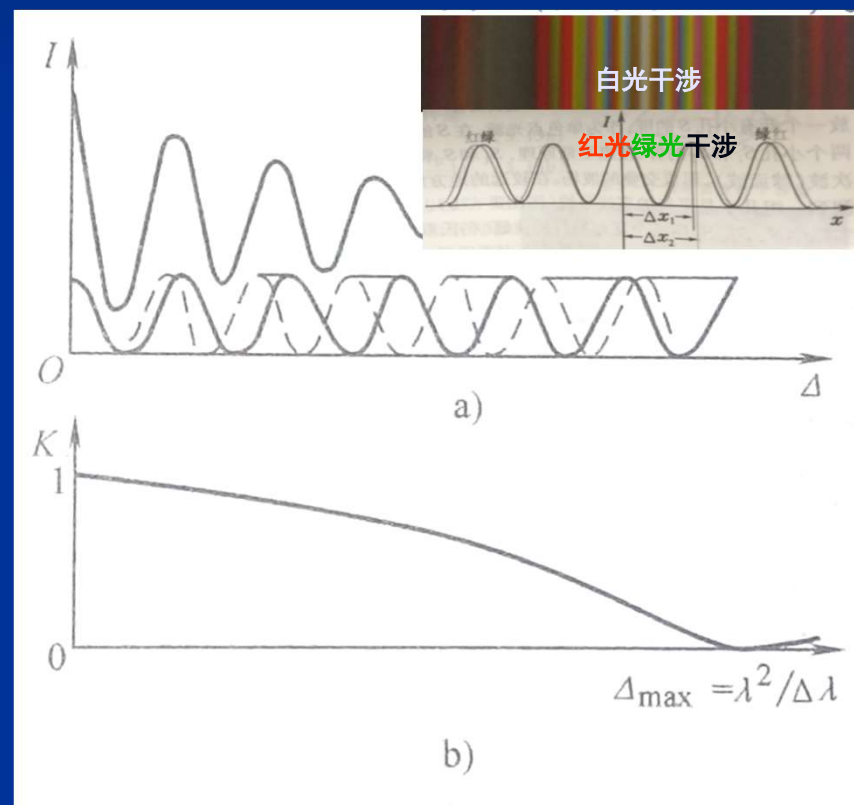
$$\theta_1 = \theta_2 = \vartheta$$



3) 波谱相干性

$$K = \left| \frac{\sin \left[(\Delta k) \cdot \frac{\Delta}{2} \right]}{(\Delta k) \cdot \frac{\Delta}{2}} \right|$$

$$(\Delta k) \cdot \frac{\Delta}{2} = 2\pi \text{ 时, } \Delta_{\max} = \frac{2\pi}{(\Delta k)} = \frac{\lambda^2}{(\Delta \lambda)}$$



4) 时间相干性

时间相干性：波列相干的持续时间

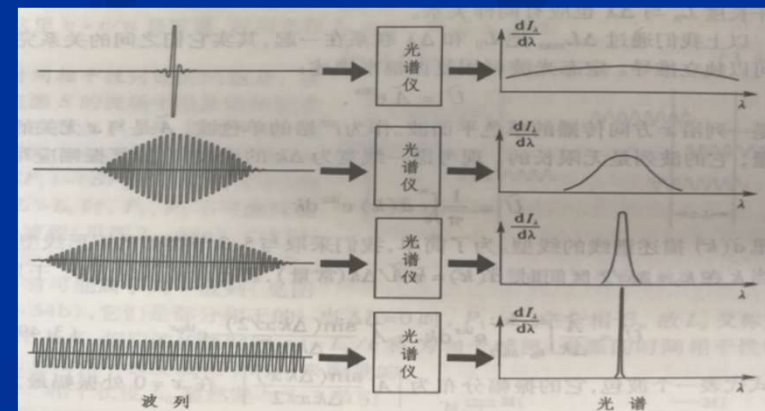
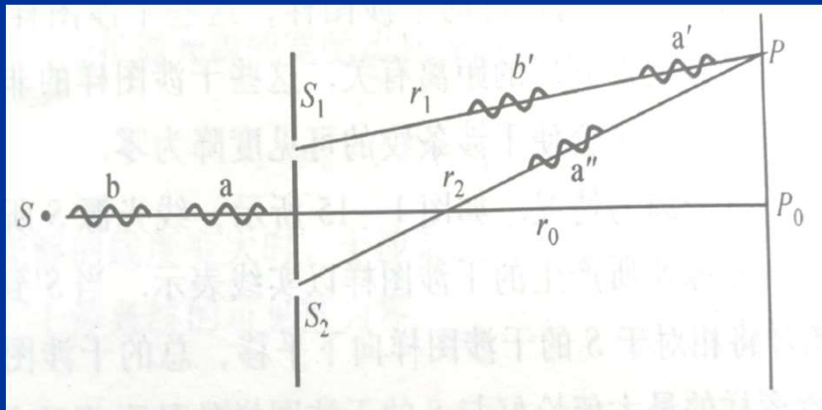
$$(\Delta t) \cdot (\Delta \nu) = 1$$

Δt 越大 $\rightarrow$ $\Delta \nu$ 越小：时间相干性越好

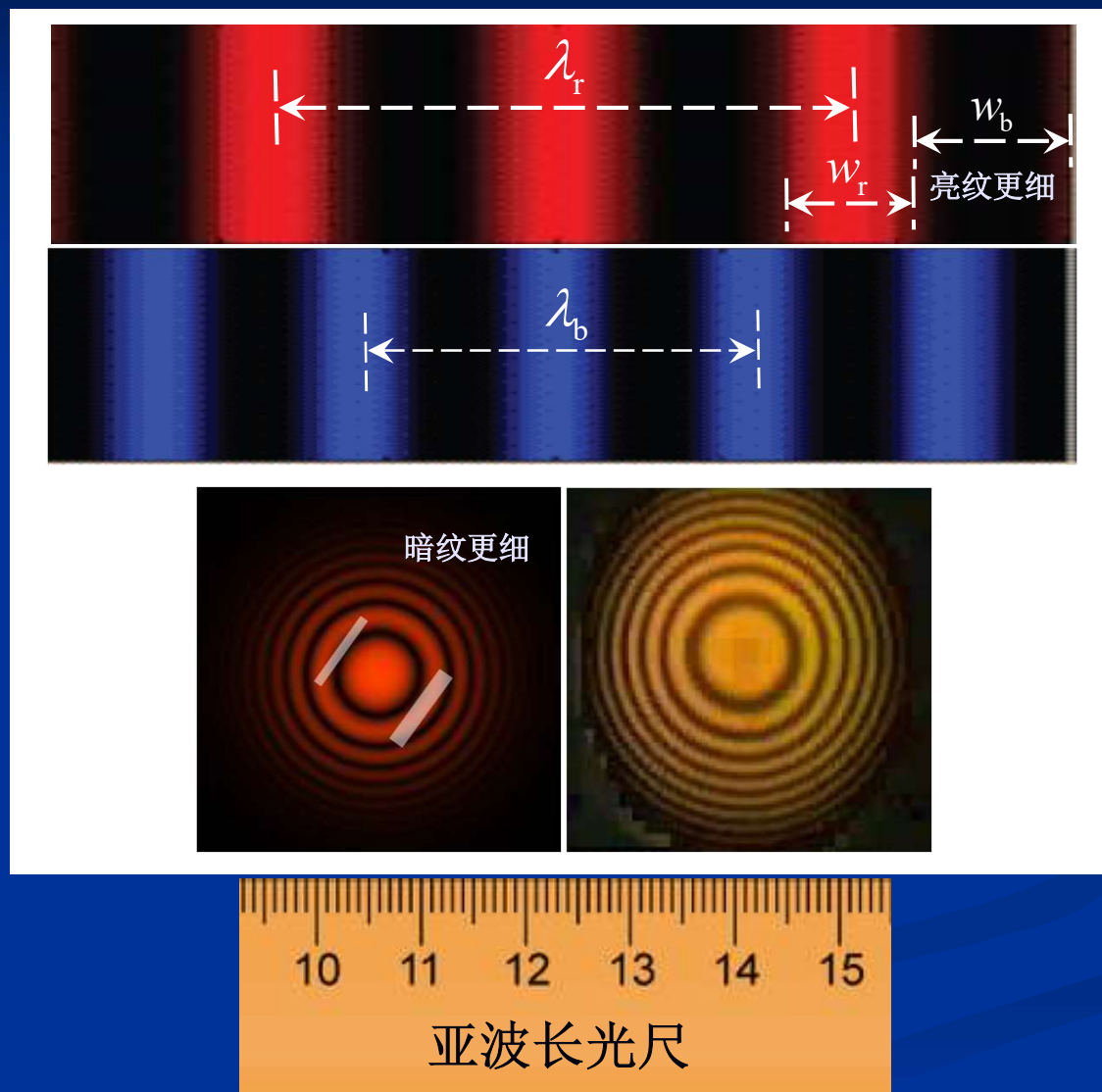
$\Delta \nu$ 越小 $\rightarrow$ 带宽越小：颜色越纯

Δt 越小 $\rightarrow$ $\Delta \nu$ 越大，带宽越大：时间相干性越差

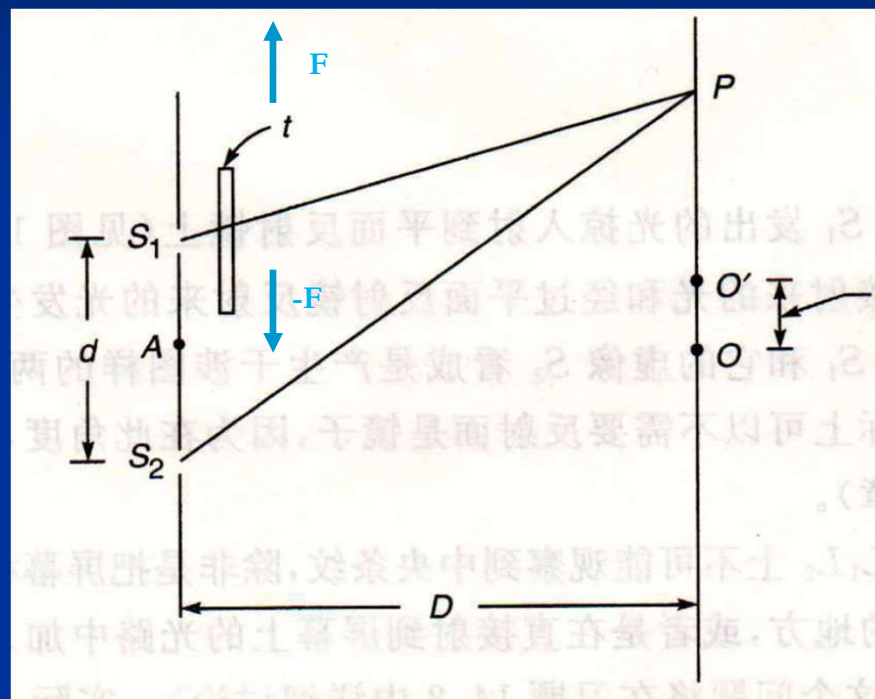
$\Delta \nu$ 越大 $\rightarrow$ 颜色可变动范围越大：多色光、甚至白光



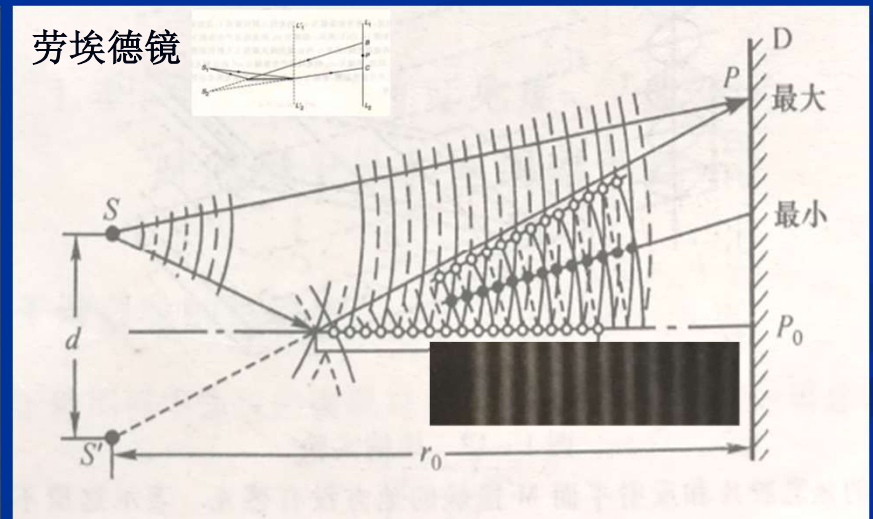
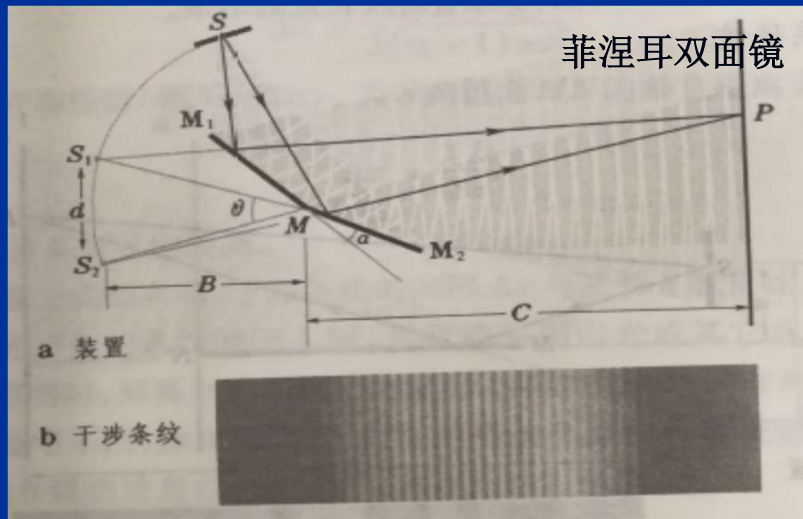
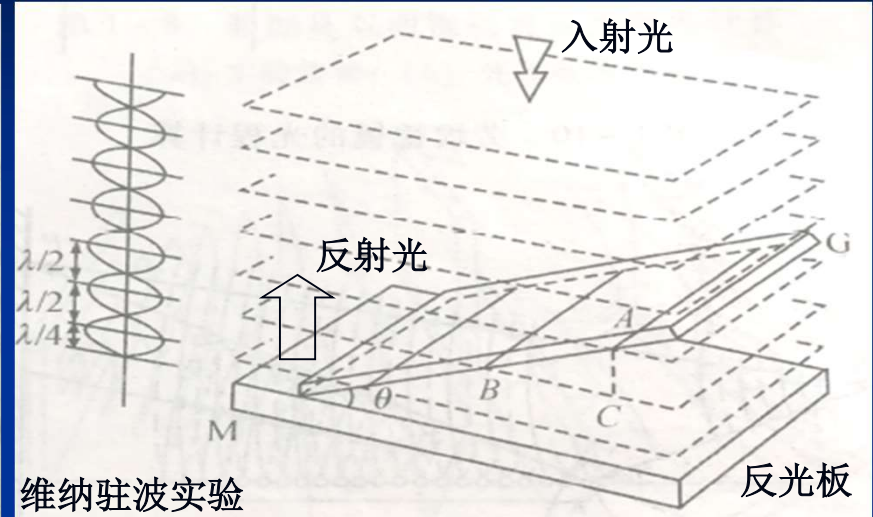
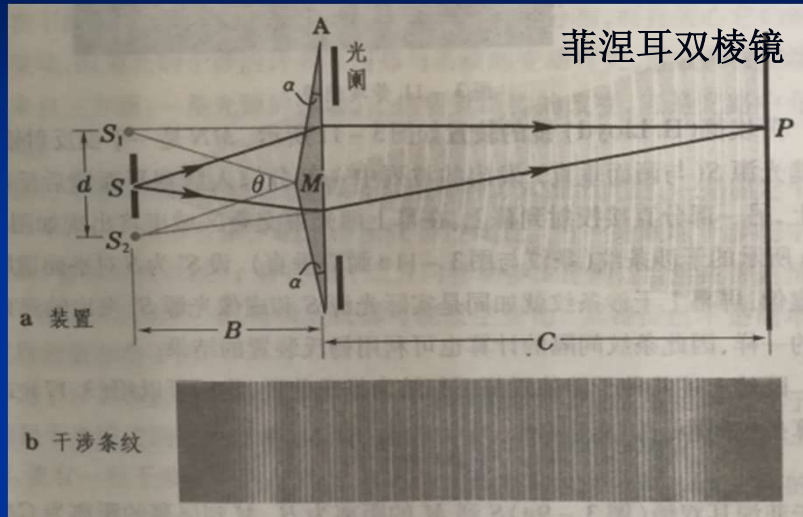
5) 亚波长精细结构



6) 移动干涉条纹



单光源典型实现方式



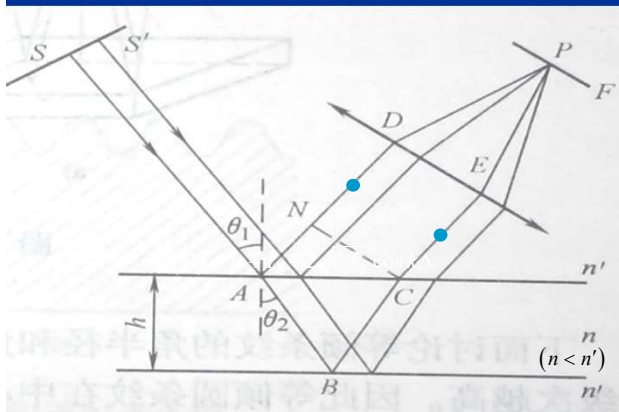
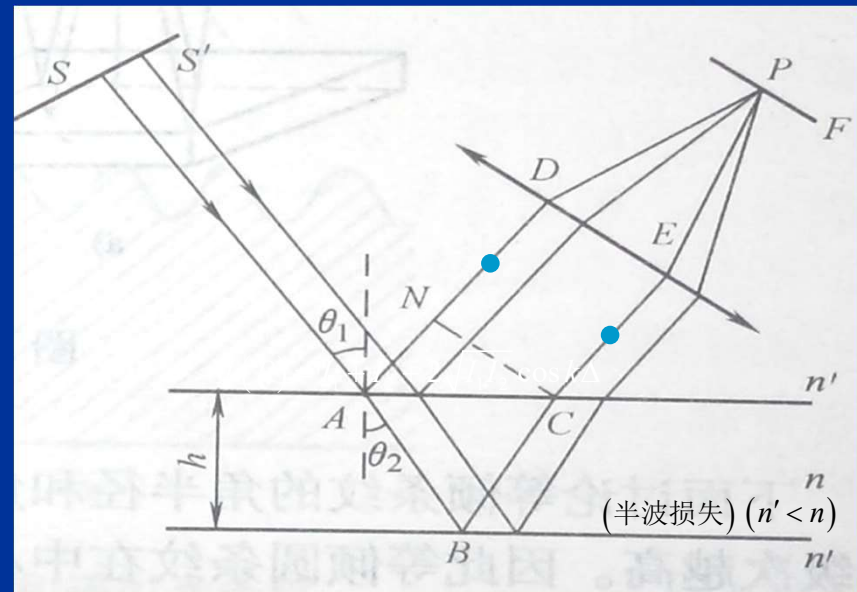
9.4 平板的双光束干涉



干涉：1) 非定域
2) 定域，空间相干性

等倾反射干涉原理

$$I(P) = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos k\Delta$$
$$\Delta = 2nh \cos \theta_2 + \frac{\lambda}{2}$$



$$\Delta = 2nh \cos \theta_2$$

中心干涉线

$$\Delta = 2nh + \frac{\lambda}{2} = m_0 \lambda = (m_1 + q) \lambda$$

m_0 : 中心亮纹干涉级

m_1 : 中心亮纹整数干涉级

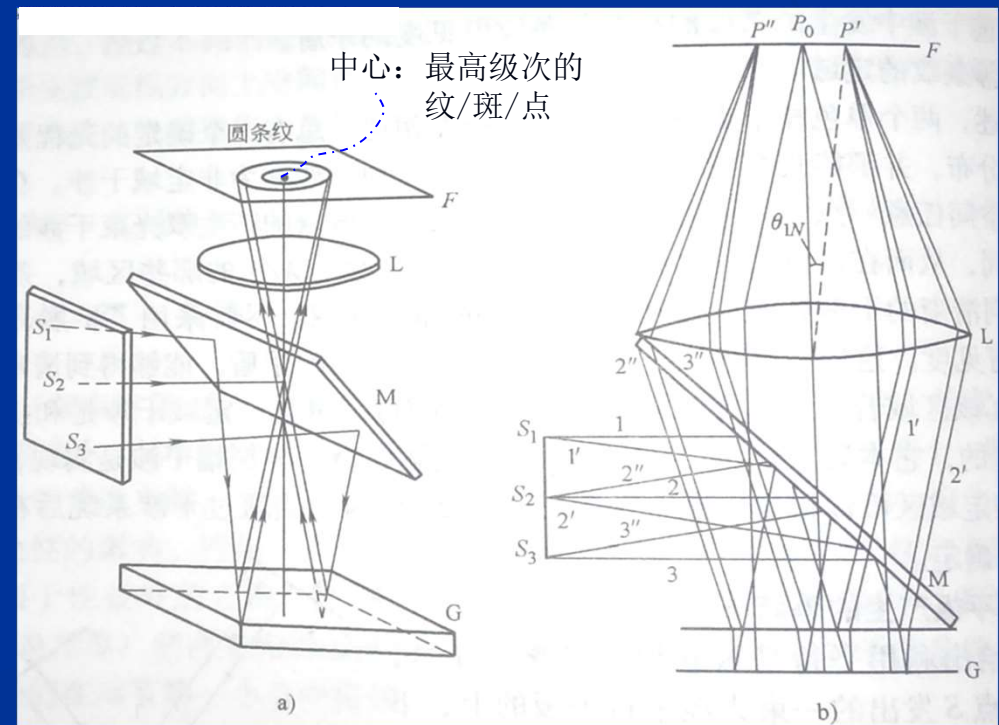
q : 小于1的分数

典型特征: 中心条纹稀疏
边缘条纹稠密
平板厚, 条纹密

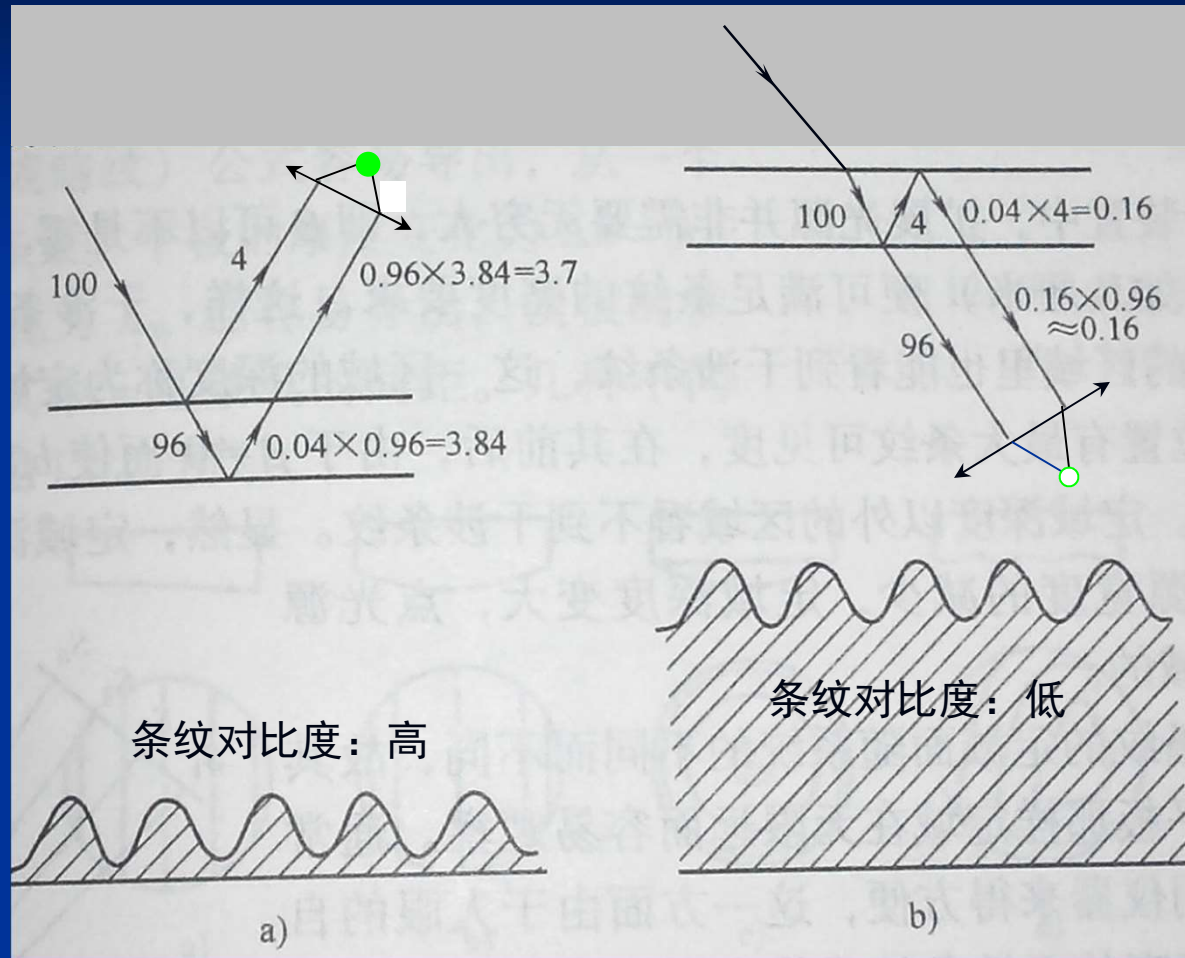
从中心亮纹向外第N个亮纹的干涉级

$$[m_1 - (N - 1)]$$

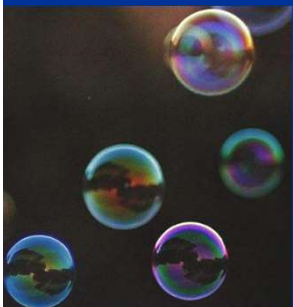
$$g_{1N} \approx \frac{1}{n'} \sqrt{\frac{n\lambda}{h}} \sqrt{N - 1 + q}$$



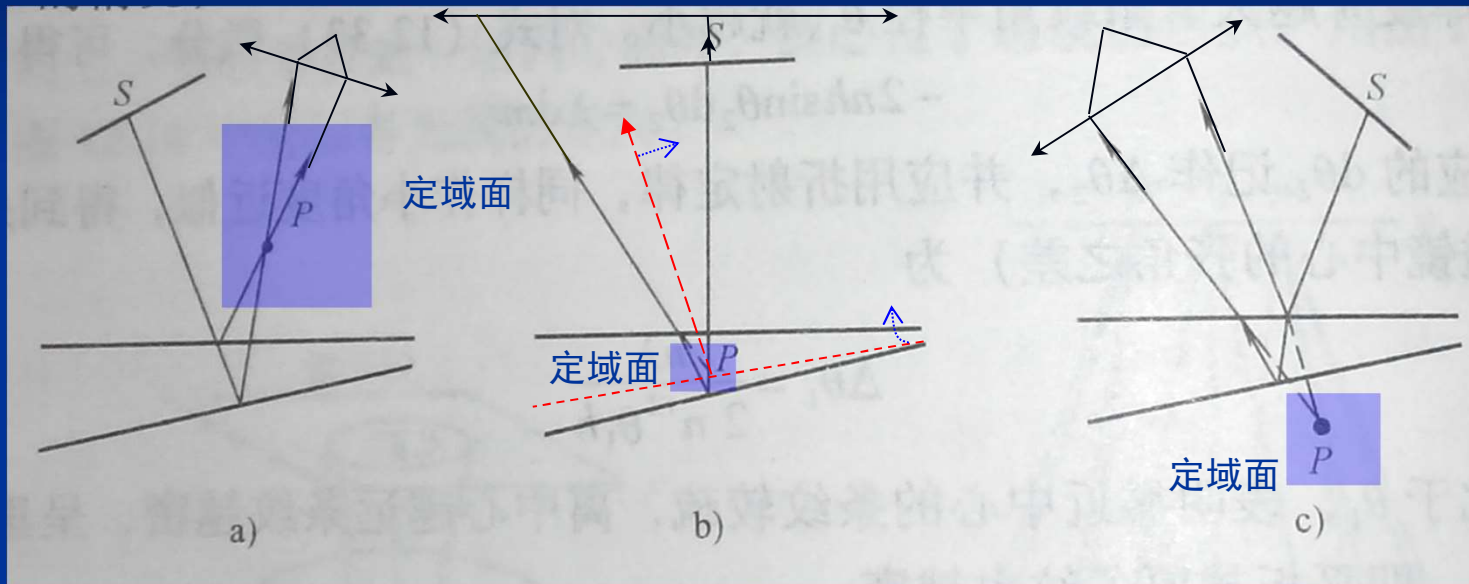
等倾透射干涉原理



无半波损失：条纹亮暗与反射光相反（能量守恒）



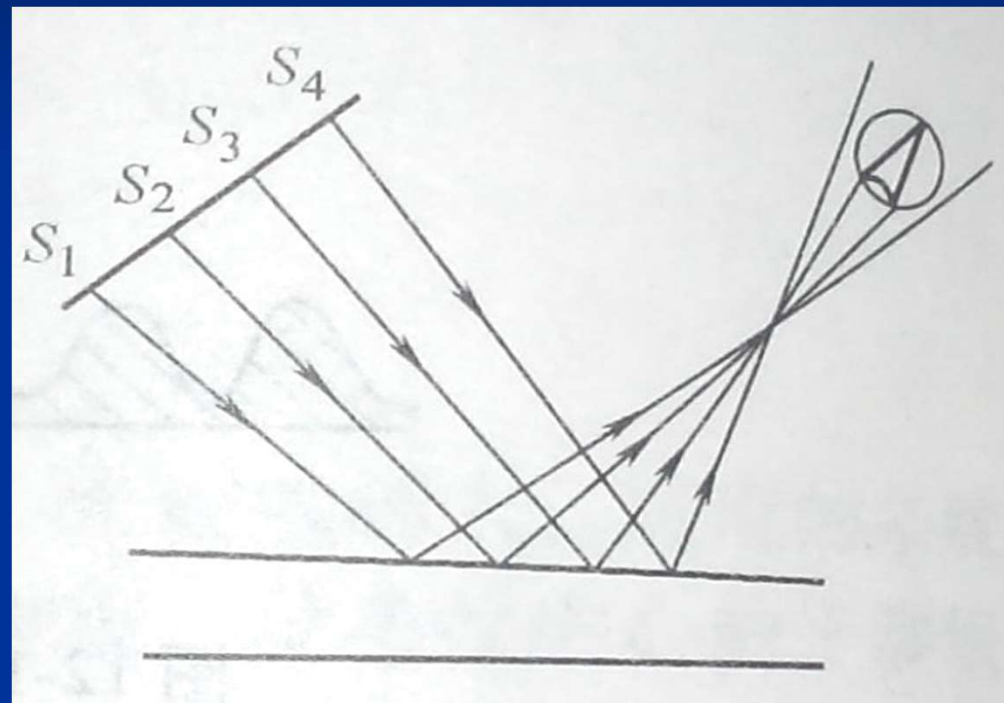
等厚干涉原理



典型特征：光源位于楔板不同部位，定域面不同。
厚度越小，定域面越接近板面。
定域深度与光源宽度成反比



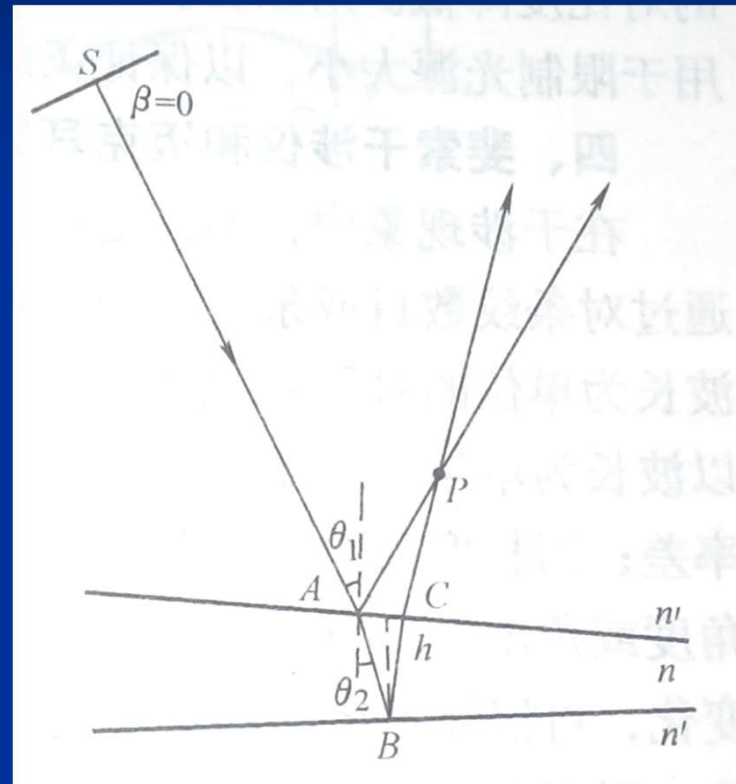
典型特征



人眼的较小瞳孔缩小了光源的实际有效尺寸
更易于找到干涉条纹

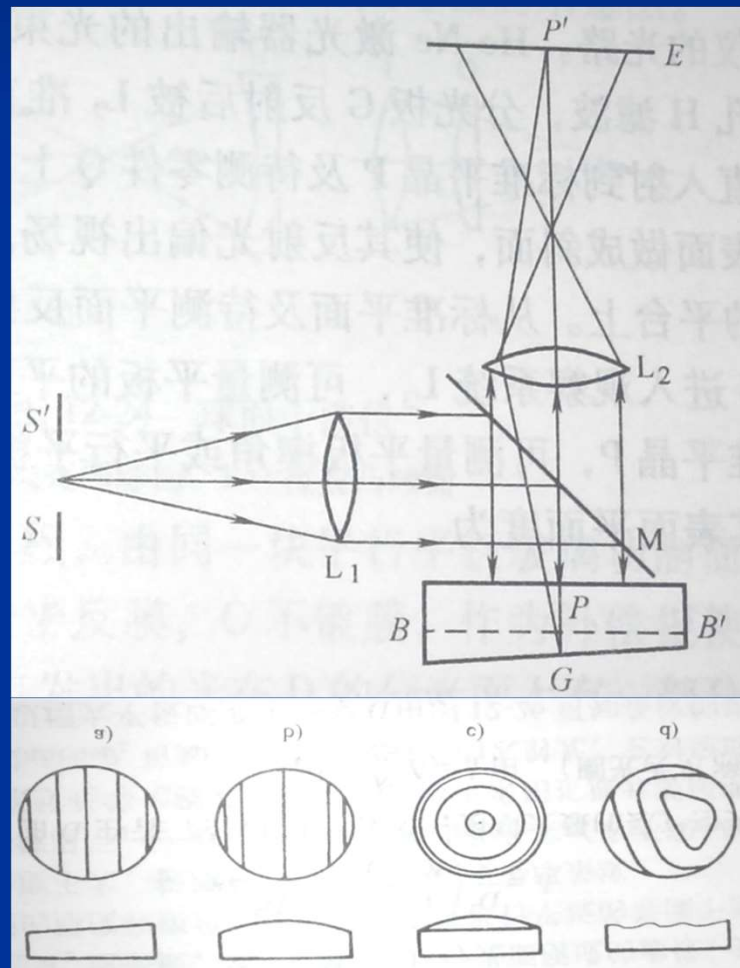
若：楔角小，楔板波

$$\Delta = n(AB + BC) - n'(AP - CP) \\ = 2nh \cos \theta_2 + \frac{\lambda}{2}$$



亮纹: $\Delta = 2nh + \frac{\lambda}{2} = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$

暗纹: $\Delta = 2nh + \frac{\lambda}{2} = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$

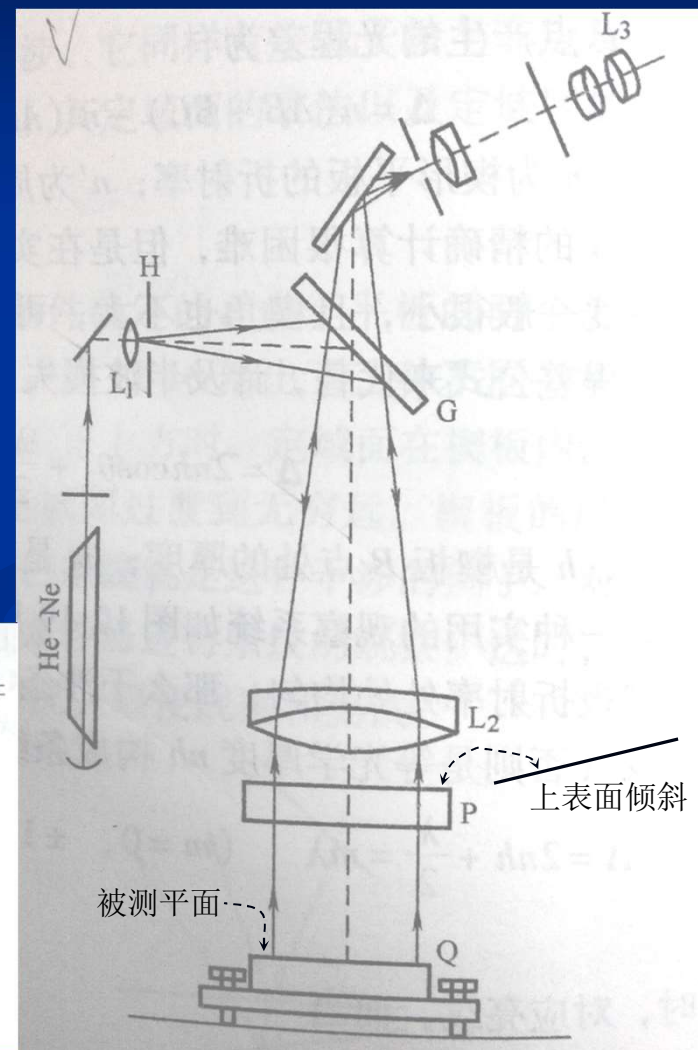
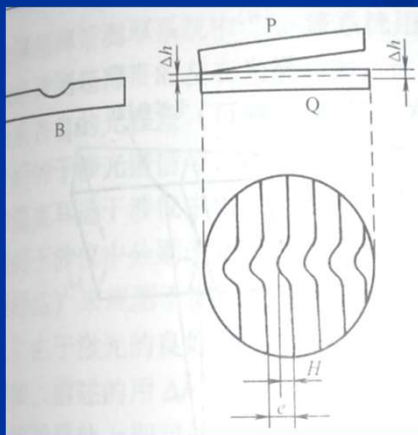


(裴索) 激光平面干涉仪

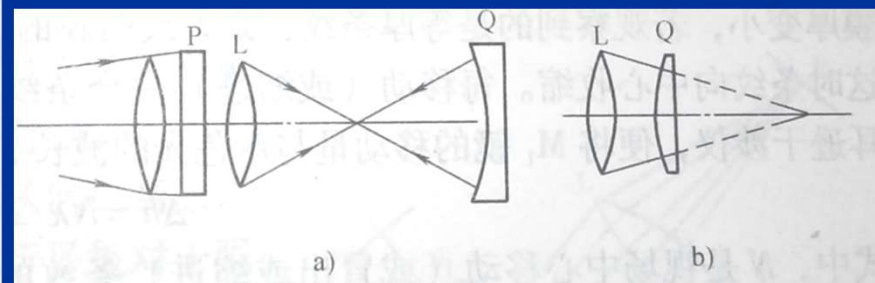
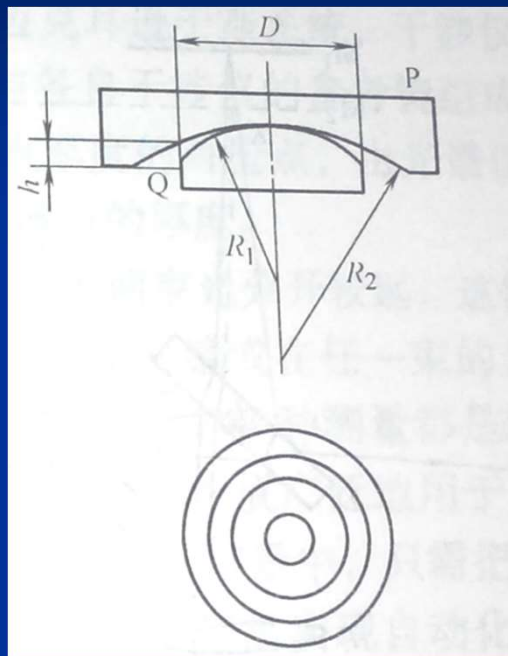
$$\Delta h = \frac{H}{e} \frac{\lambda}{2}$$

H : 被测表面平整度偏差引起的条纹偏离

e : 条纹间距



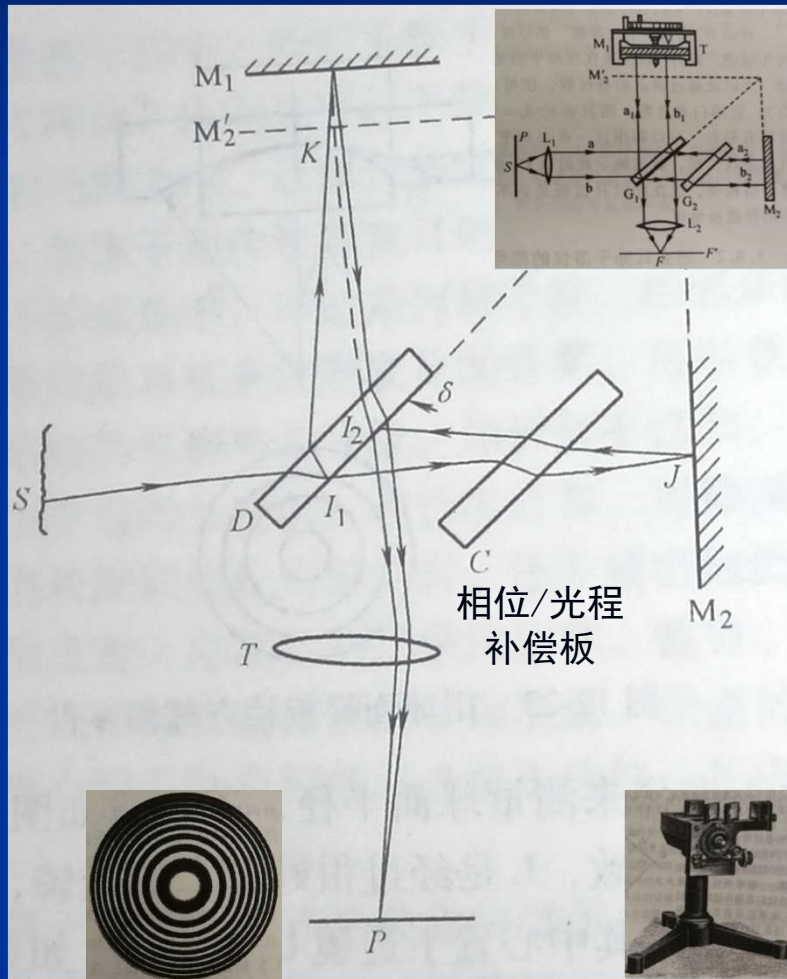
(裴索) 激光球面干涉仪



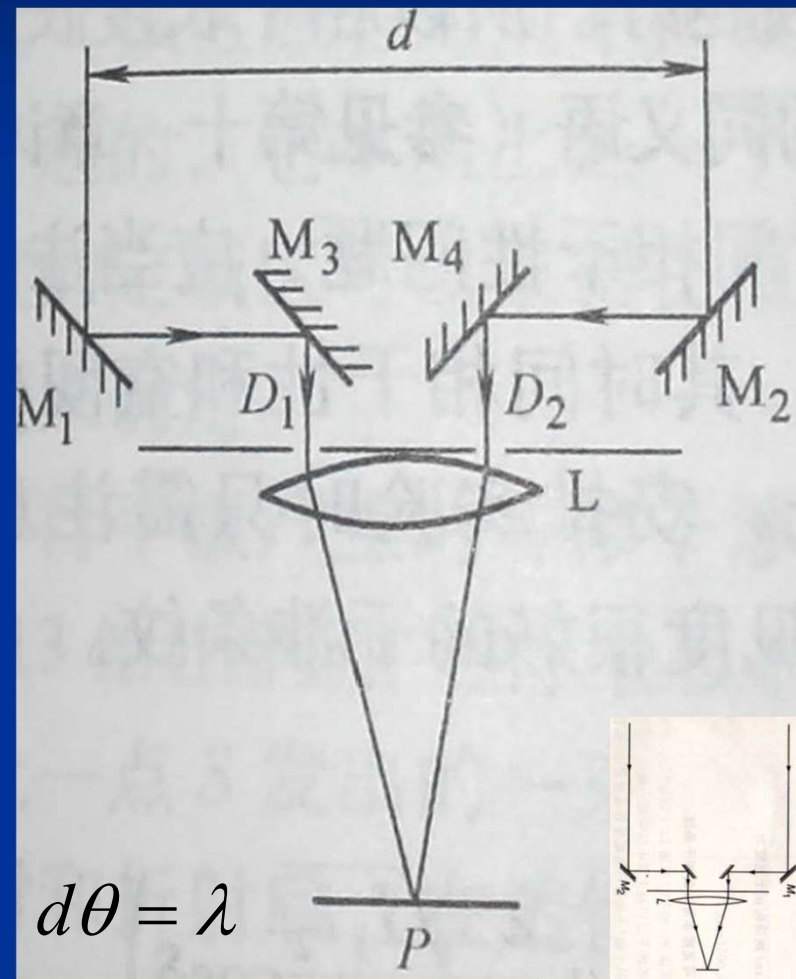
$$h = \frac{D^2}{8} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{D^2}{8} \Delta k = N \frac{\lambda}{2}$$

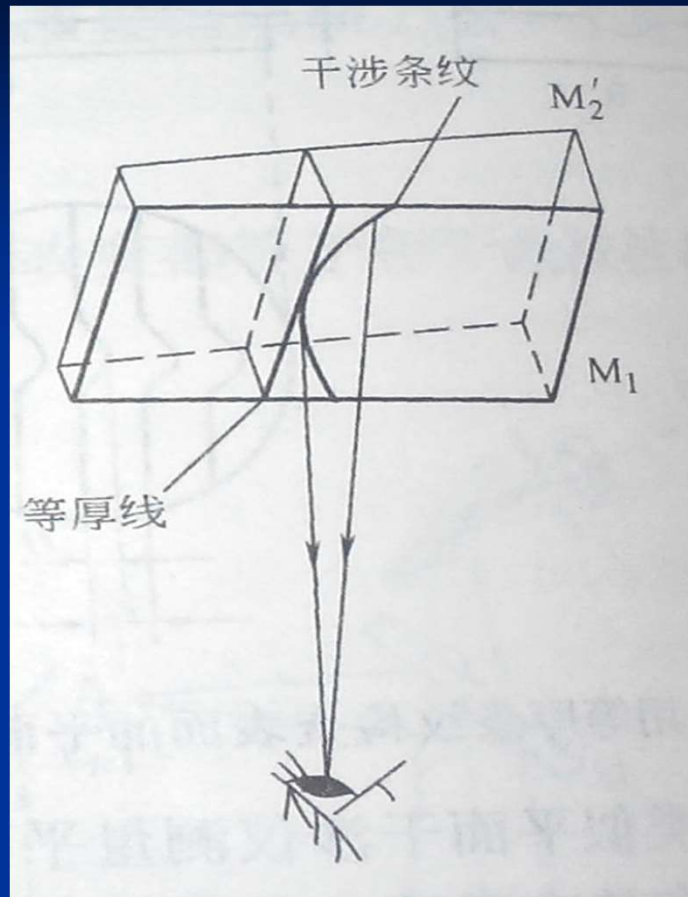
有： $N = \frac{D^2}{4\lambda} \Delta k$, 即为 D 范围内观察到 N 个圆条纹，光圈为 N

迈克尔逊干涉仪



迈克耳逊测星干涉仪



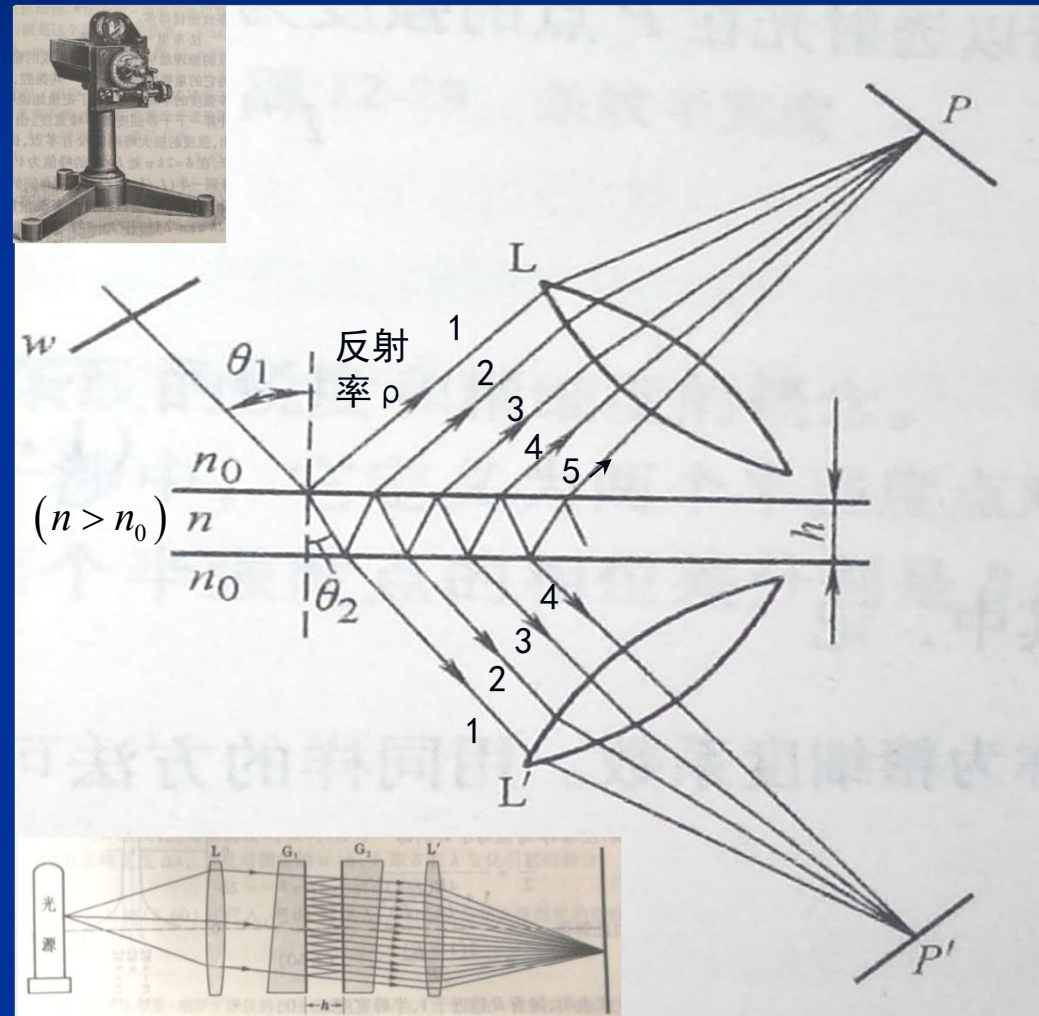


光程差: $\Delta = 2nh \cos \vartheta$

ϑ 大, h 应增大, 或 n 增大

干涉条纹弯向楔顶

9.5 平行平板的多光束干涉及其应用



1) 数值分析

若 $I_0=1$, $\rho=0.9$, $n>n_0$, 有:

$$\begin{aligned} I_1 &= 0.9, & I_2 &= 0.009, & I_3 &= 0.0073, & I_4 &= 0.00577, \\ & & & & & & I_5 &= 0.0046, \dots \\ A^{(r)}_1 &= -0.948683, & A^{(r)}_2 &= 0.094868, & A^{(r)}_3 &= 0.085440, & A^{(r)}_4 &= 0.075961, \\ & & & & & & A^{(r)}_5 &= 0.067823, \dots \end{aligned}$$

$$\begin{aligned} A^{(r)} &= \sum_i A_i^{(r)} = -0.948683 + (0.094868 + 0.08544 + 0.075961 + 0.067823 + \dots) \\ &= -0.948683 + (\sim 0.1 + \sim 0.1 + \sim 0.1 + \sim 0.1 + \dots) \\ &= -0.948683 + (\sim 0.9) \rightarrow 0 \end{aligned}$$

$$I_{\text{反射}} = A^{(r)2} = A^{(r)} \cdot A^{(r)*} \rightarrow 0, \quad \text{即: 入射光几乎全部透射, } I_{\text{透射}} \rightarrow 1$$

2) 理论分析

若 $I_0=1$, $r_{\text{振幅}} \rightarrow \text{介质/板}$, $r'_{\text{振幅}} \rightarrow \text{板/介质}$,
 $t_{\text{振幅}} \rightarrow \text{介质/板}$, $t'_{\text{振幅}} \rightarrow \text{板/介质}$,
 $n > n_0$,

有:

$$\begin{aligned} A^{(t)}_1 &= tt' A_0, & A^{(t)}_2 &= tt' r'^2 A_0, & A^{(t)}_3 &= tt' r'^4 A_0, & A^{(t)}_4 &= tt' r'^6 A_0, & \dots \\ & & & & & & A^{(t)}_p &= tt' r'^{2(p-1)} A_0, & \dots \\ \tilde{A}^{(t)}_1 &= tt' A_0, & \tilde{A}^{(t)}_2 &= tt' r'^2 e^{j\delta} A_0, & \tilde{A}^{(t)}_3 &= tt' r'^4 e^{j2\delta} A_0, & \tilde{A}^{(t)}_4 &= tt' r'^6 e^{j3\delta} A_0, & \dots \\ & & & & & & \tilde{A}^{(t)}_p &= tt' r'^{2(p-1)} e^{j(p-1)\delta} A_0, & \dots \end{aligned}$$

透射复振幅为:

$$\begin{aligned} \tilde{A}^{(t)} &= \sum_i \tilde{A}^{(t)}_i = tt' A_0 \left(1 + r'^2 e^{j\delta} + r'^4 e^{j2\delta} + r'^6 e^{j3\delta} + \dots + r'^{2(p-1)} e^{j(p-1)\delta} + \dots \right) \\ &= \frac{tt'}{1 - r'^2 e^{j\delta}} A_0 \end{aligned} \quad \delta = \frac{4\pi}{\lambda} nh \cos \theta_2$$

考虑基本关系: $r = -r', tt' = 1 - r^2, \rho_{\text{界面反射比}} = r^2, \tau_{\text{界面透射比}} = 1 - \rho$

透射光强为:

$$I^{(t)} = \tilde{A}^{(t)} \cdot \tilde{A}^{(t)*} = \frac{\tau^2}{(1 - \rho)^2 + 4\rho \sin^2 \frac{\delta}{2}} = \frac{1}{1 + F \sin^2 \frac{\delta}{2}}$$

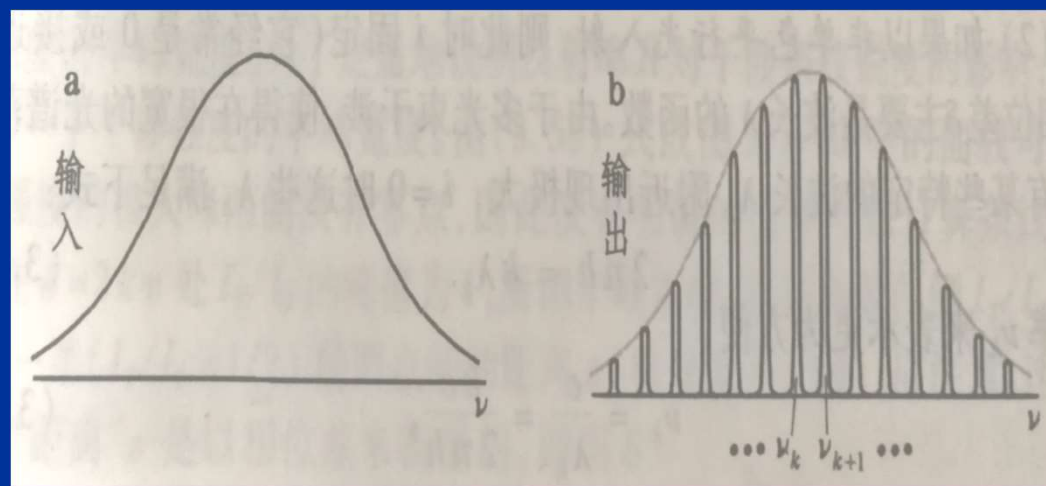
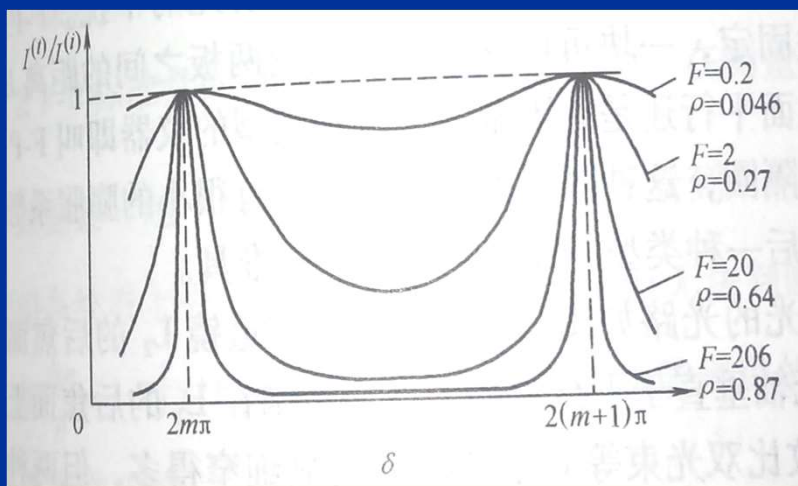
反射光强为:

$$I^{(r)} = 1 - I^{(t)} = \frac{F \sin^2 \frac{\delta}{2}}{1 + F \sin^2 \frac{\delta}{2}} I_0$$

$$\text{精细度系数: } F = \frac{4\rho}{(1 - \rho)^2}$$

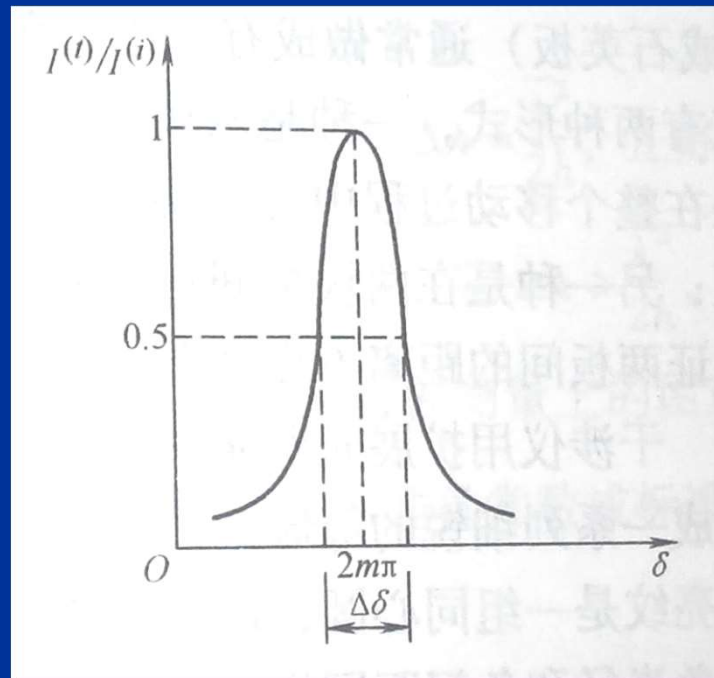
基本特征:

- 1) 等倾干涉
- 2) ρ 越大 $\rightarrow F$ 越大 $\rightarrow$ 干涉条纹精细度越高
- 3) 垂直平板观察, 等倾圆环
- 4) 精细选频

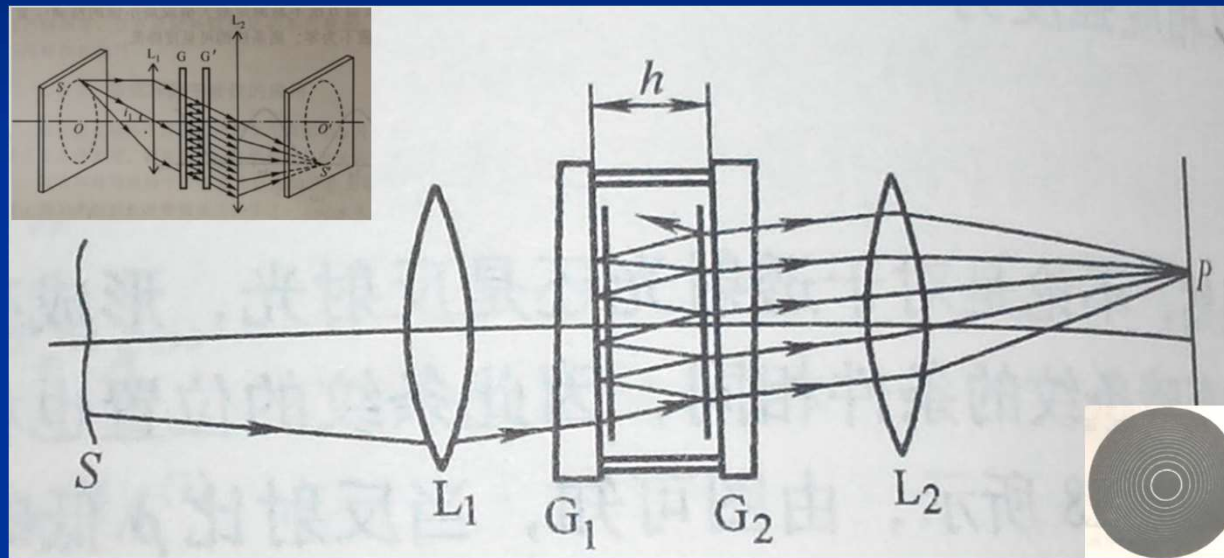


条纹锐度/条纹的相位半宽度: $\Delta\delta = \frac{4}{\sqrt{F}} = \frac{2(1-\rho)}{\sqrt{\rho}}$

条纹精细度: $s = \frac{2\pi}{\Delta\delta} = \frac{\pi\sqrt{F}}{2} = \frac{\pi\sqrt{\rho}}{1-\rho}$



Febry-Perot (FP) 干涉腔/具/仪



基本特征:

- 1) G_1 、 G_2 上的高反膜的表面平整度: $1/20 \sim 1/100\lambda$ 级;
- 2) G_1 、 G_2 保持一小楔角, 将高反膜外的结构上的反射光投射出去;
- 3) 单色性极好, 高光谱;
- 4) 调变 h , 调变干涉光波的波长.

$$\Delta m = m_1 - m_2 = \left(\frac{2h}{\lambda_1} + \frac{\varphi}{\pi} \right) - \left(\frac{2h}{\lambda_2} + \frac{\varphi}{\pi} \right) = \frac{2h(\lambda_2 - \lambda_1)}{\lambda_1 \lambda_2} = \frac{\Delta e}{e}$$

$$\Delta \lambda = \lambda_2 - \lambda_1 = \left(\frac{\Delta e}{e} \right) \frac{\bar{\lambda}^2}{2h}$$

标准具常数/标准具自由光谱范围:

$$(\Delta \lambda)_{(S,R)} = \frac{\bar{\lambda}^2}{2h}, \quad \frac{\Delta e}{e} = 1$$

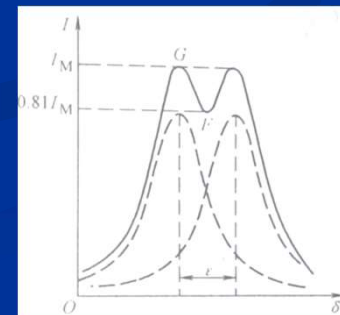
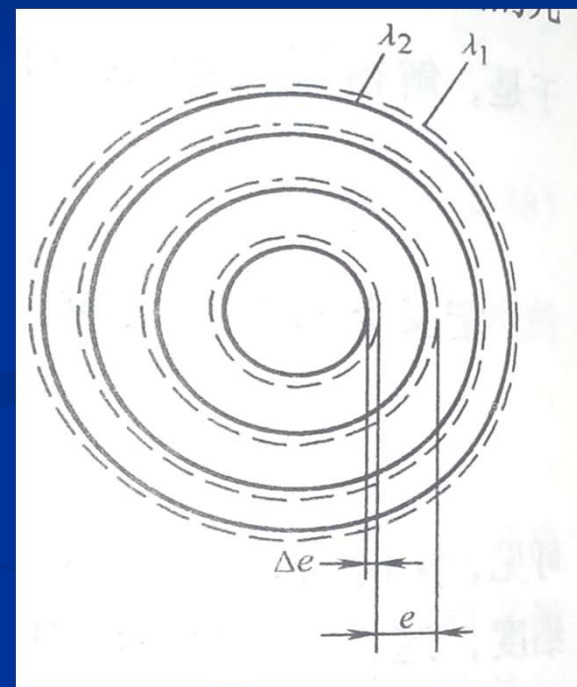
标准具分辨极限/可分辨的最小波长差:

$$(\Delta \lambda)_{\min}$$

标准具分辨本领:

$$A = \frac{\lambda}{(\Delta \lambda)_{\min}} = mN = 0.97ms$$

采用瑞利判据

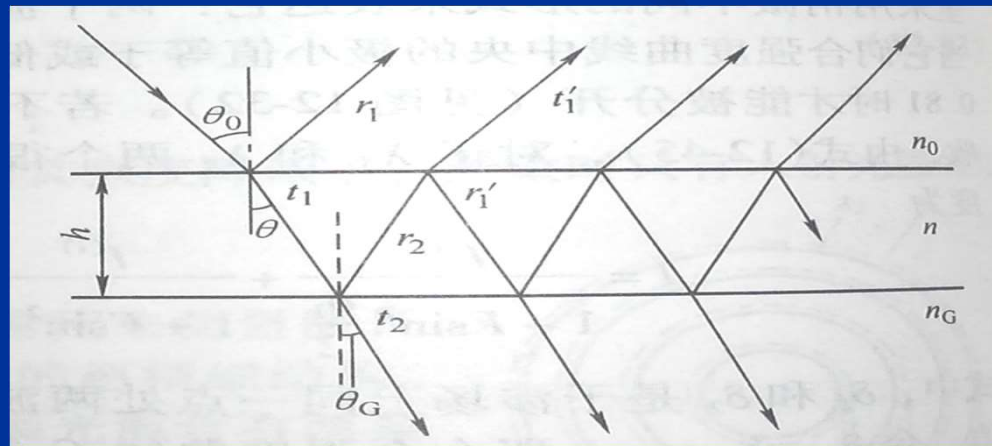


Febry-Perot (FP) 干涉膜

$$A^{(r)} = \frac{r_1 + r_2 e^{j\delta}}{1 + r_1 r_2 e^{j\delta}} A_0 \quad A^{(t)} = \frac{t_1 t_2}{1 + r_1 r_2 e^{j\delta}} A_0 \quad \delta = \frac{4\pi}{\lambda} n h \cos \theta$$

$$\rho = |r|^2 = \frac{r_1^2 + r_2^2 + 2r_1 r_2 \cos \delta}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos \delta}$$

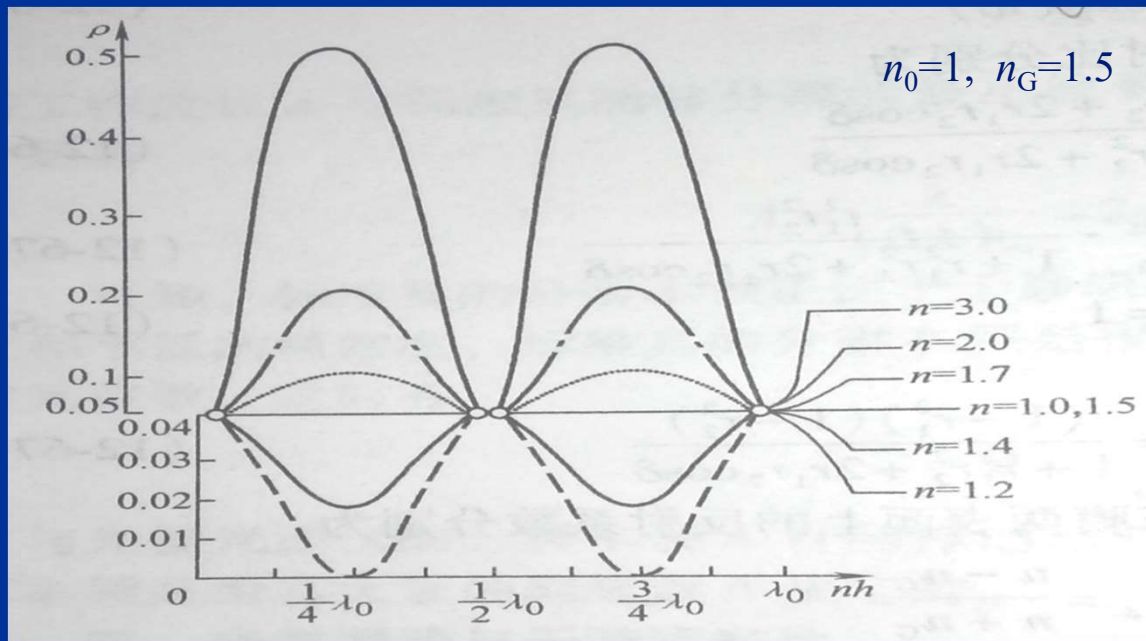
$$\tau = \frac{n_G \cos \theta_G}{n_0 \cos \theta_0} \frac{t_1^2 t_2^2}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos \delta} = \frac{(1 - r_1^2)(1 - r_2^2)}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos \delta}$$



1) 正入射

$$r_1 = \frac{n_0 - n}{n_0 + n} \quad r_2 = \frac{n - n_G}{n + n_G}$$

$$\rho = \frac{(n_0 - n_G)^2 \cos^2 \frac{\delta}{2} + \left(\frac{n_0 n_G}{n} - n \right)^2 \sin^2 \frac{\delta}{2}}{(n_0 + n_G)^2 \cos^2 \frac{\delta}{2} + \left(\frac{n_0 n_G}{n} + n \right)^2 \sin^2 \frac{\delta}{2}}$$

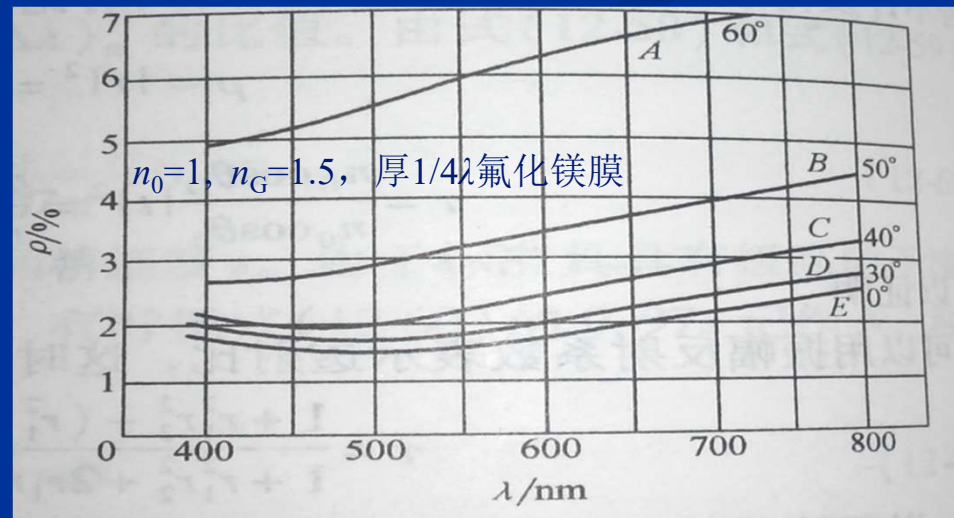


2) 正入射单层增透膜

1) 若 $n < n_G$, 则有:

$$\rho = \frac{\left(\frac{n_0 n_G}{n} - n \right)^2}{\left(\frac{n_0 n_G}{n} + n \right)^2} = \left[\frac{n_0 - \frac{n^2}{n_G}}{n_0 + \frac{n^2}{n_G}} \right]^2$$

2) 若 $n = \sqrt{n_0 n_G}$, 则 $\rho = 0$, 全透



3) 斜入射单层增透膜

等效折射率

$$r_s = \frac{n_2 \cos \theta_2 - n_1 \cos \theta_1}{n_2 \cos \theta_2 + n_1 \cos \theta_1} = \frac{\bar{n}_2 - \bar{n}_1}{\bar{n}_2 + \bar{n}_1}$$

$$r_p = \frac{n_2 / \cos \theta_2 - n_1 / \cos \theta_1}{n_2 / \cos \theta_2 + n_1 / \cos \theta_1}$$

4) 单层增反膜

1) 若 $n > n_G$, $nh = \frac{\lambda}{4}$ 时, 则有:

$$\rho_{\lambda 0} = \left[\frac{n_0 - \frac{n^2}{n_G}}{n_0 + \frac{n^2}{n_G}} \right]^2$$

n 越大, 膜系反射比越高.

5) 半波长膜/虚膜

若 $nh = \frac{\lambda}{2}$, 膜系反射比同无膜层情形

6) 双层膜

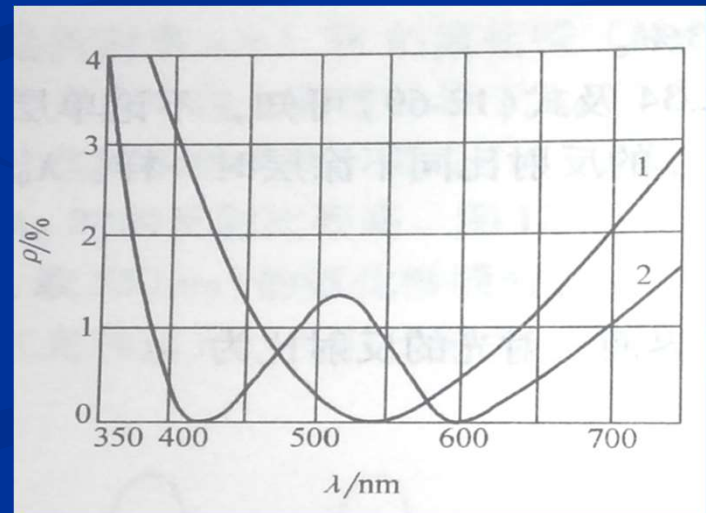
垂直 $n_0 - n_G$ 界面入射光: $\rho = \frac{\left(\frac{n_0 n_G}{n} - n\right)^2}{\left(\frac{n_0 n_G}{n} + n\right)^2} = \left[\frac{n_0 - \frac{n^2}{n_G}}{n_0 + \frac{n^2}{n_G}} \right]^2$ 等效界面折射率
 $= \left[\frac{n_0 - n'}{n_0 + n'} \right]^2$ (一个界面)

垂直 $n_0 - n_1 - n_2 - n_G$ 界面入射光: $n'_{G2} = \frac{n_2^2}{n_G}, n'_{G1} = \frac{n_1^2}{n'_{G2}} n_G$

$$\rho = \left(\frac{n_0 - n'_{G1}}{n_0 + n'_{G1}} \right)^2$$

基本特征:

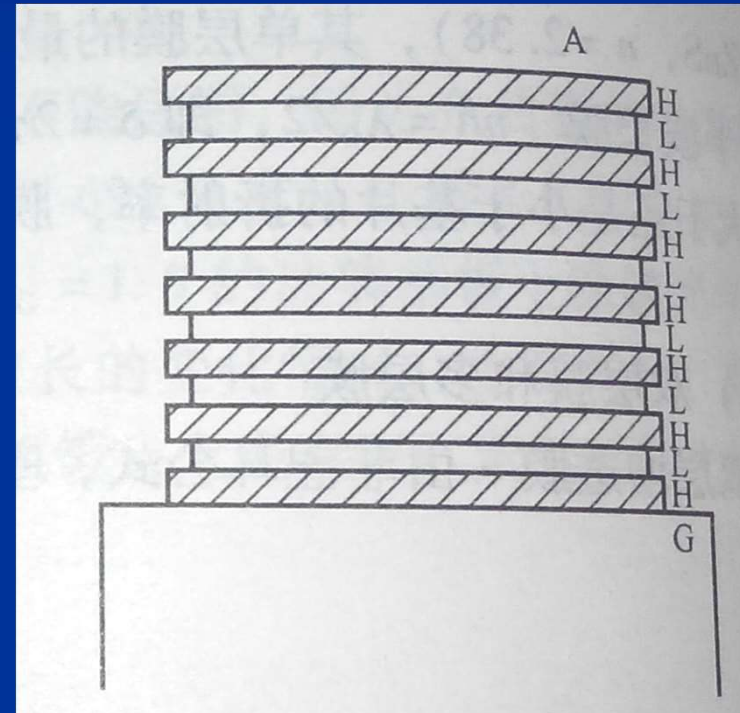
- 1) 针对特征波长有效
- 2) 普遍情形较单层膜差



7) 多层膜

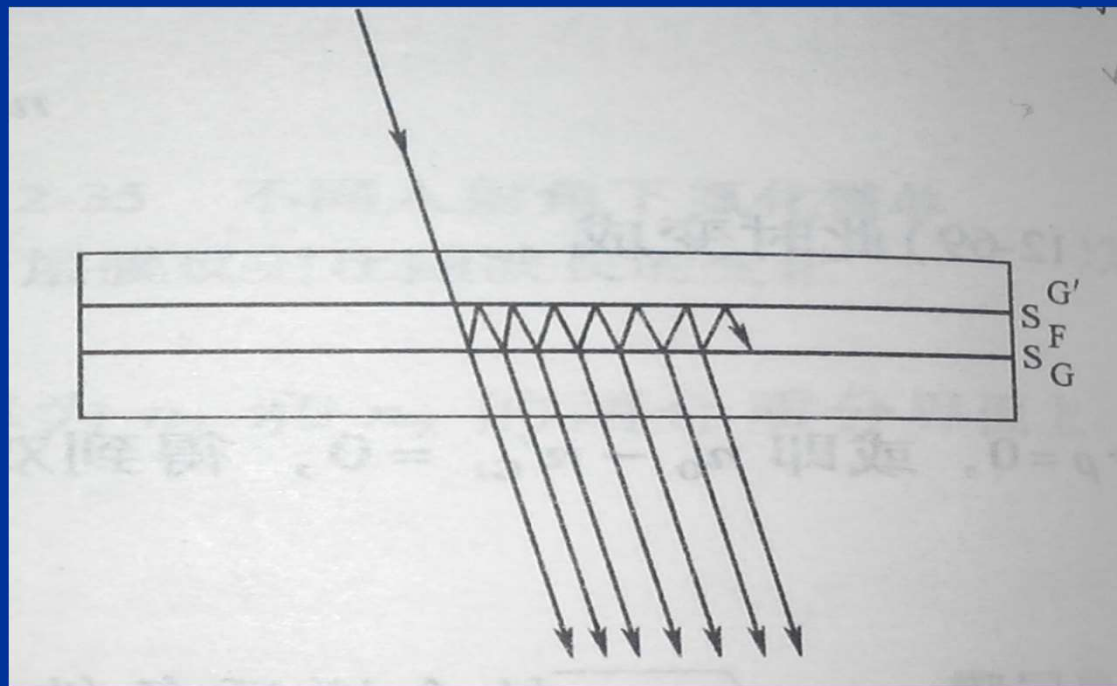
基本特征：常为 nh 均为 $1/4\lambda$ 的高折射、低折射膜交替叠合而成
 $GHLHLH...GHA=G(HL)^pHA$, $p=1, 2, 3, \dots$

$$\rho = \left[\frac{n_0 - \left(\frac{n_H}{n_L} \right)^{2p} \frac{n_H^2}{n_G}}{n_0 + \left(\frac{n_H}{n_L} \right)^{2p} \frac{n_H^2}{n_G}} \right]^2$$



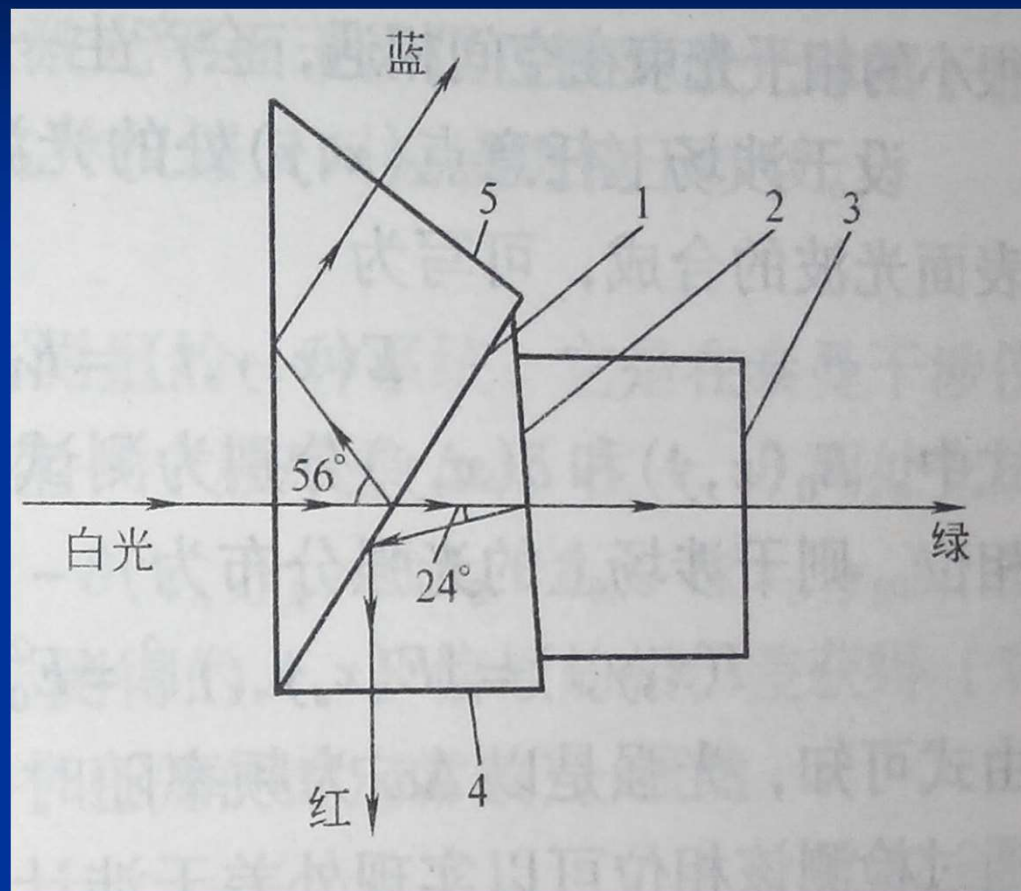
常用光学薄膜

$$\lambda_c = 2nh / m, \quad \Delta\lambda = \frac{\lambda^2}{2\pi nh} \frac{1-\rho}{\sqrt{\rho}}, \quad \tau = \frac{I^{(t)}}{I_0} \leq 1$$



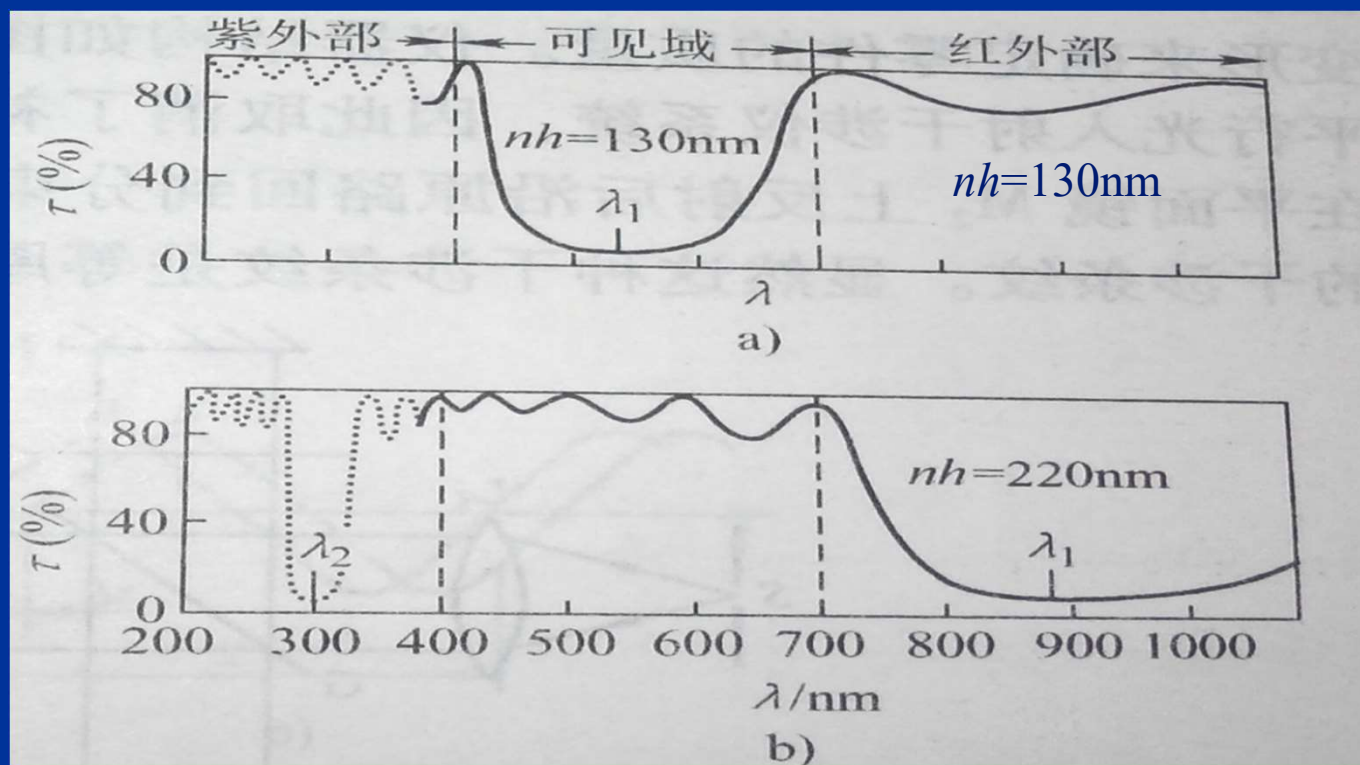
干涉滤光片

彩色分光膜



红外滤光片

$$\rho = \frac{(n_0 - n_G)^2 \cos^2 \frac{\delta}{2} + \left(\frac{n_0 n_G}{n} - n \right)^2 \sin^2 \frac{\delta}{2}}{(n_0 + n_G)^2 \cos^2 \frac{\delta}{2} + \left(\frac{n_0 n_G}{n} + n \right)^2 \sin^2 \frac{\delta}{2}}$$



9.6 现代干涉技术和干涉系统

1) 外差干涉

测试光波

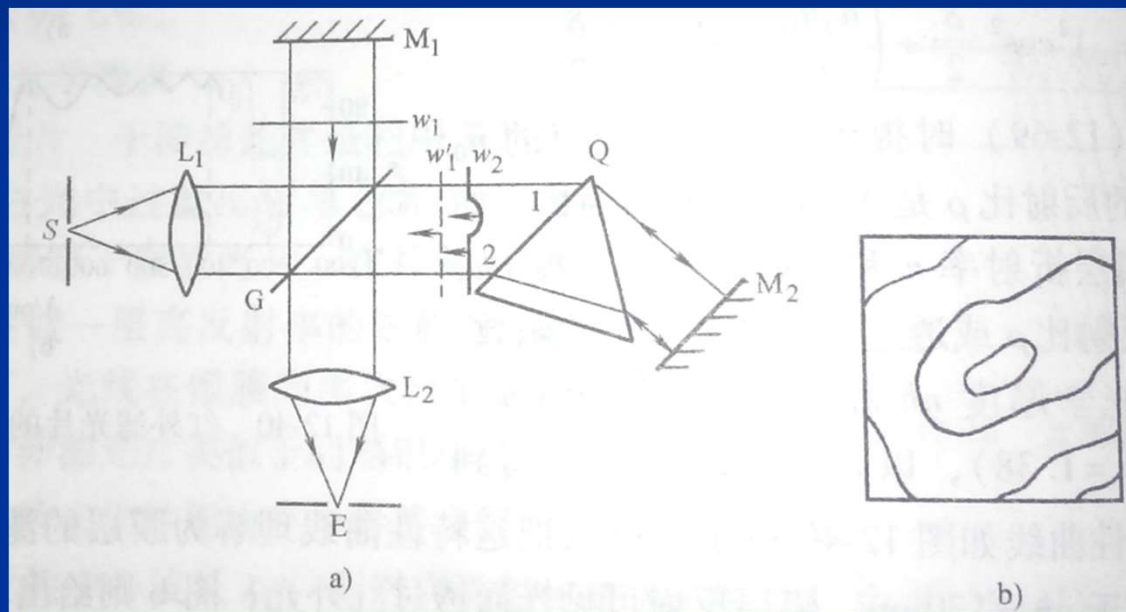
参考光波

$$E(x, y, t) = E_0(x, y) \exp[j(\delta(x, y) - \omega t)] + E_r \exp[j(\delta_r - (\omega + \Delta\omega)t)]$$

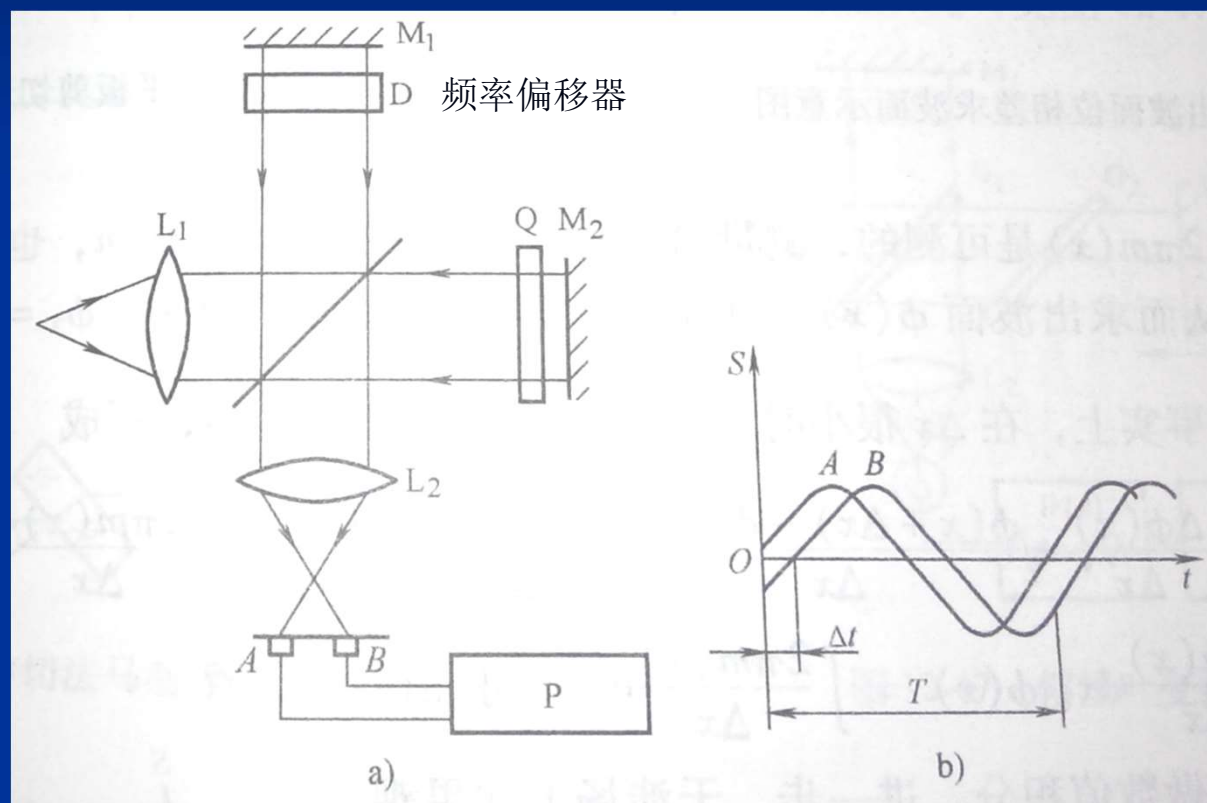
差分运算

$$I(x, y, t) = |E(x, y, t)|^2 = E_0^2(x, y) + E_r^2 + 2E_0(x, y)E_r \cos[\Delta\omega t + \delta_r - \delta(x, y)]$$

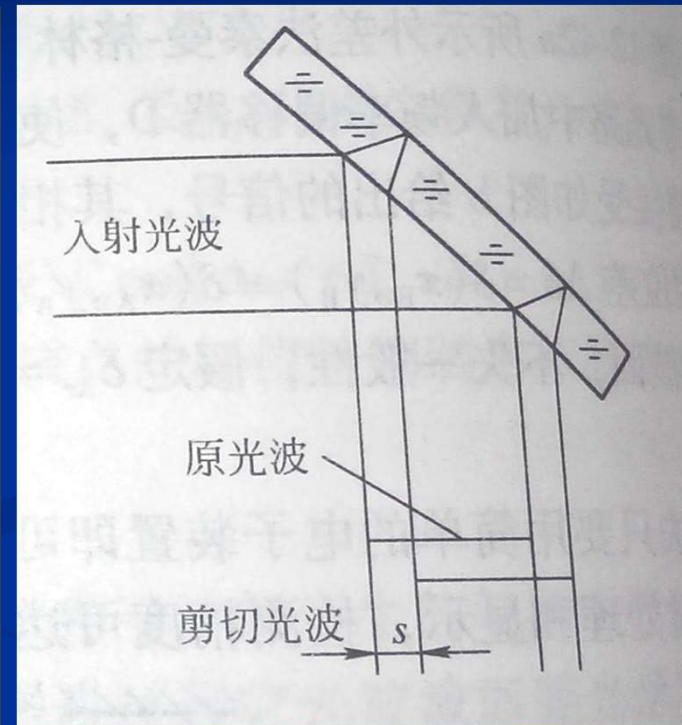
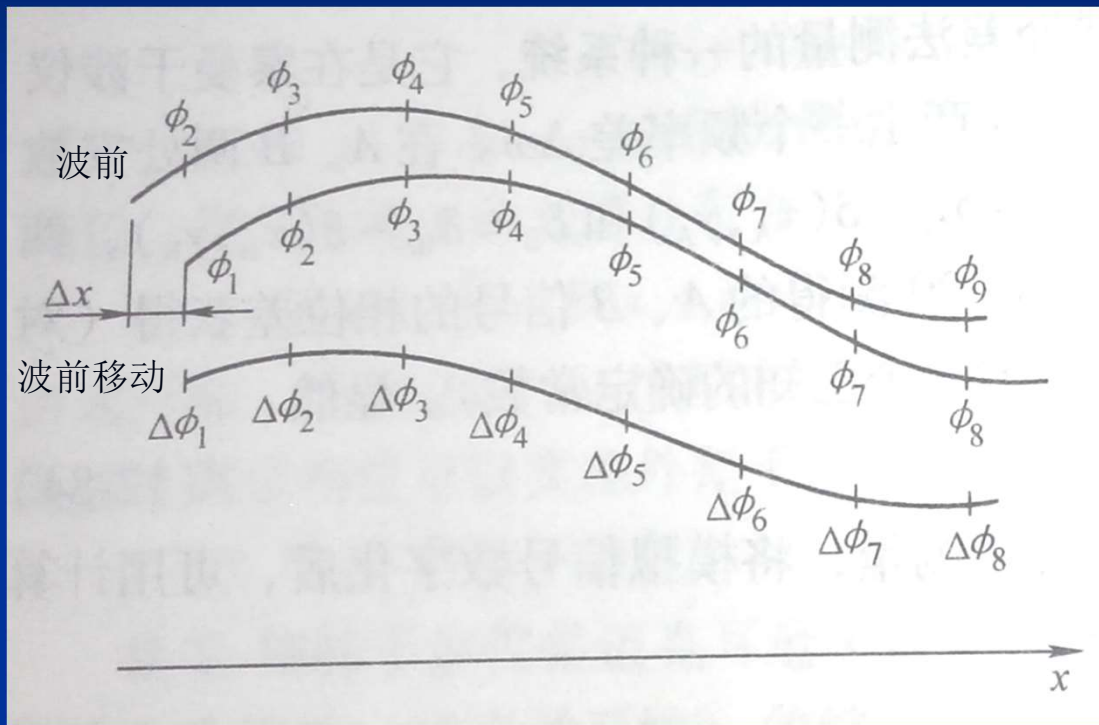
2) 泰曼-格林干涉仪



3) 外差法泰曼-格林干涉仪



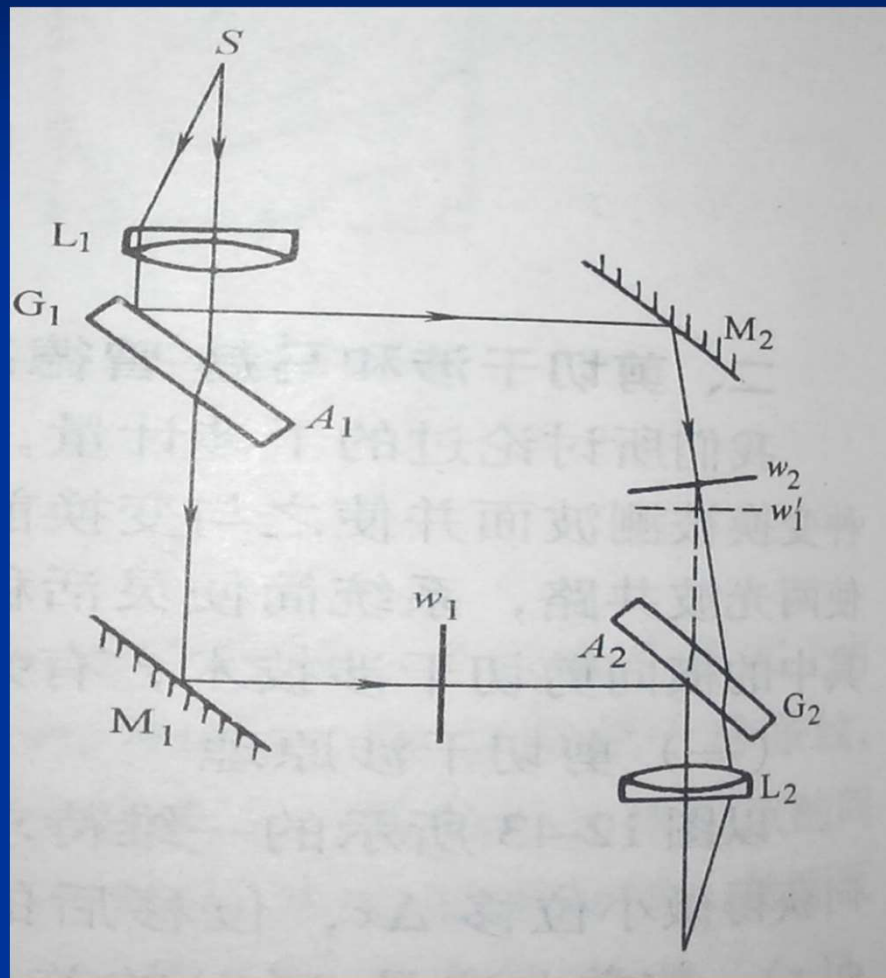
4) 剪切干涉/波前调制



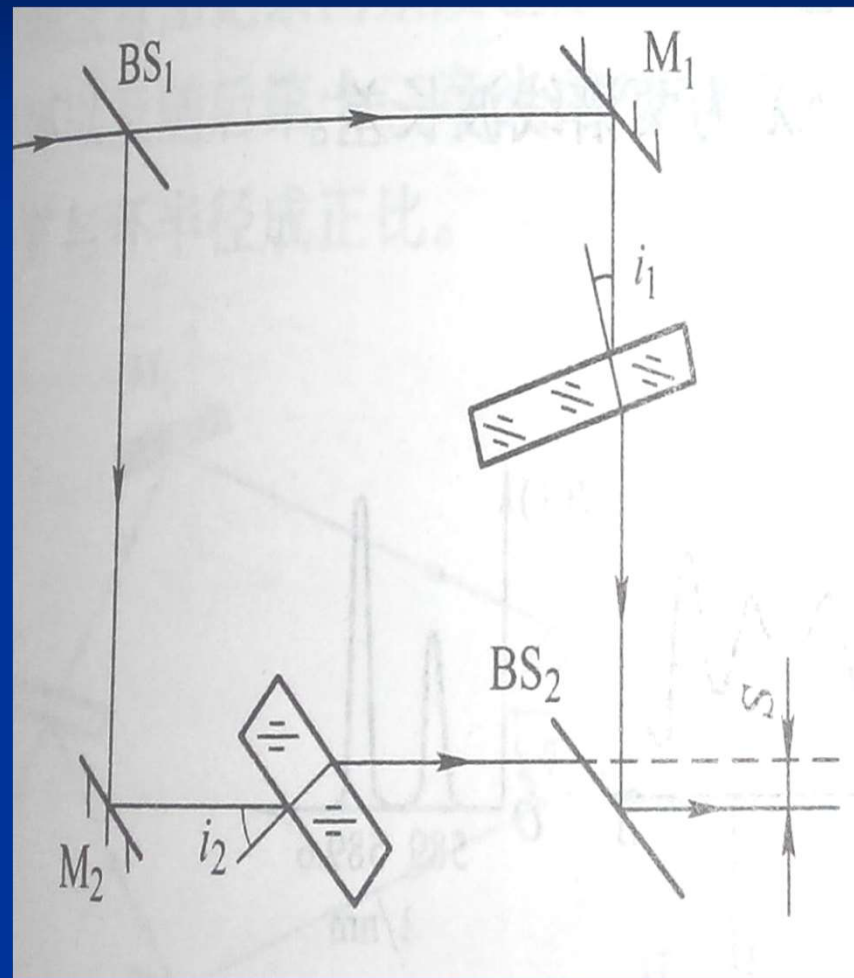
$$\Delta\varphi(x) = \varphi(x + \Delta x) - \varphi(x)$$

平板剪切干涉

5) 马赫-曾德干涉仪/波前调制

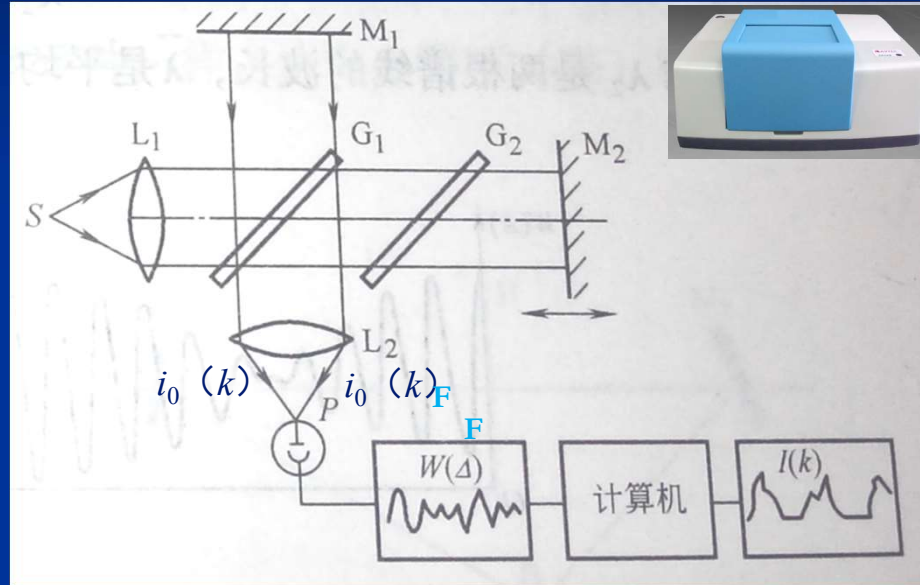


等厚干涉



剪切法

6) 傅立叶变换光谱仪/振幅调制

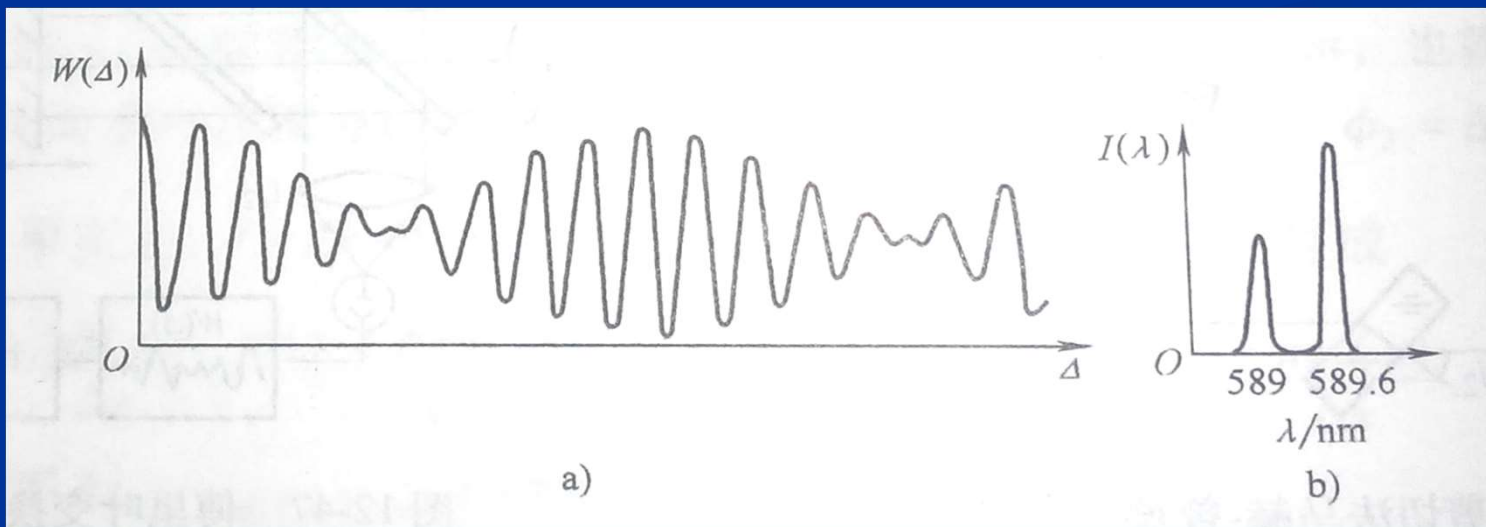


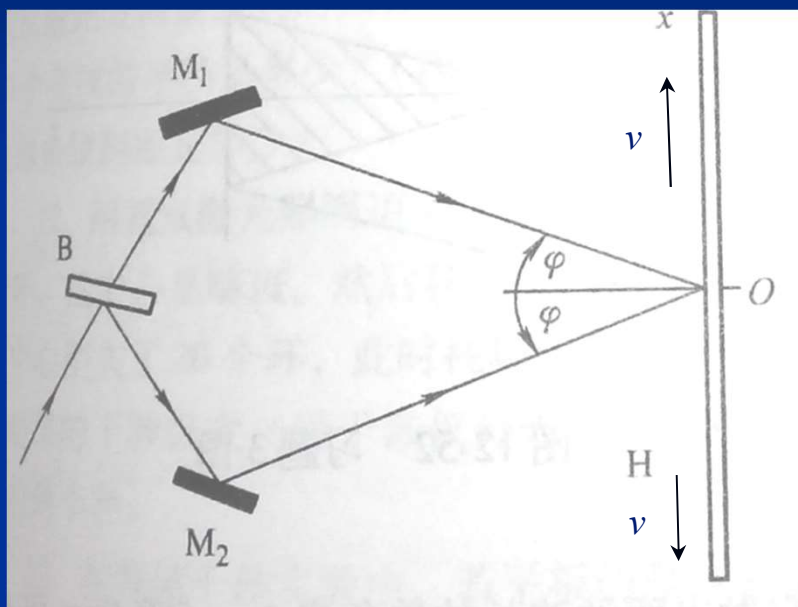
$$\begin{aligned}
 I(\Delta) &= \int_{K_0 - \frac{\Delta k}{2}}^{K_0 + \frac{\Delta k}{2}} 2i_0(1 + \cos k\Delta) dk = \int_{-\infty}^{+\infty} 2i_0(k)(1 + \cos k\Delta) dk \\
 &= \int_{-\infty}^{+\infty} 2i_0(k) dk + \int_{-\infty}^{+\infty} 2i_0(k) \frac{\exp(jk\Delta) + \exp(-jk\Delta)}{2} dk = \frac{1}{2} I(0) + \int_{-\infty}^{+\infty} i_0(k) \exp(jk\Delta) dk
 \end{aligned}$$

$$W(\Delta) = i(\Delta) - \frac{1}{2}I(0) = \int_{-\infty}^{+\infty} i_0(k) \exp(jk\Delta) dk$$

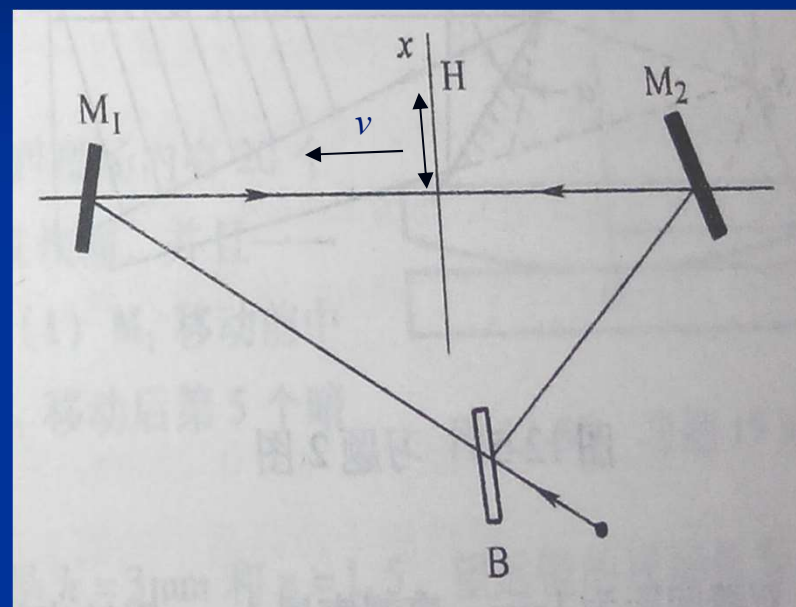
$$i_0(k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} W(\Delta) \exp(-jk\Delta) d\Delta$$

$$\Delta' = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} = \frac{\bar{\lambda}^2}{\Delta \lambda}$$





制作光栅/透镜



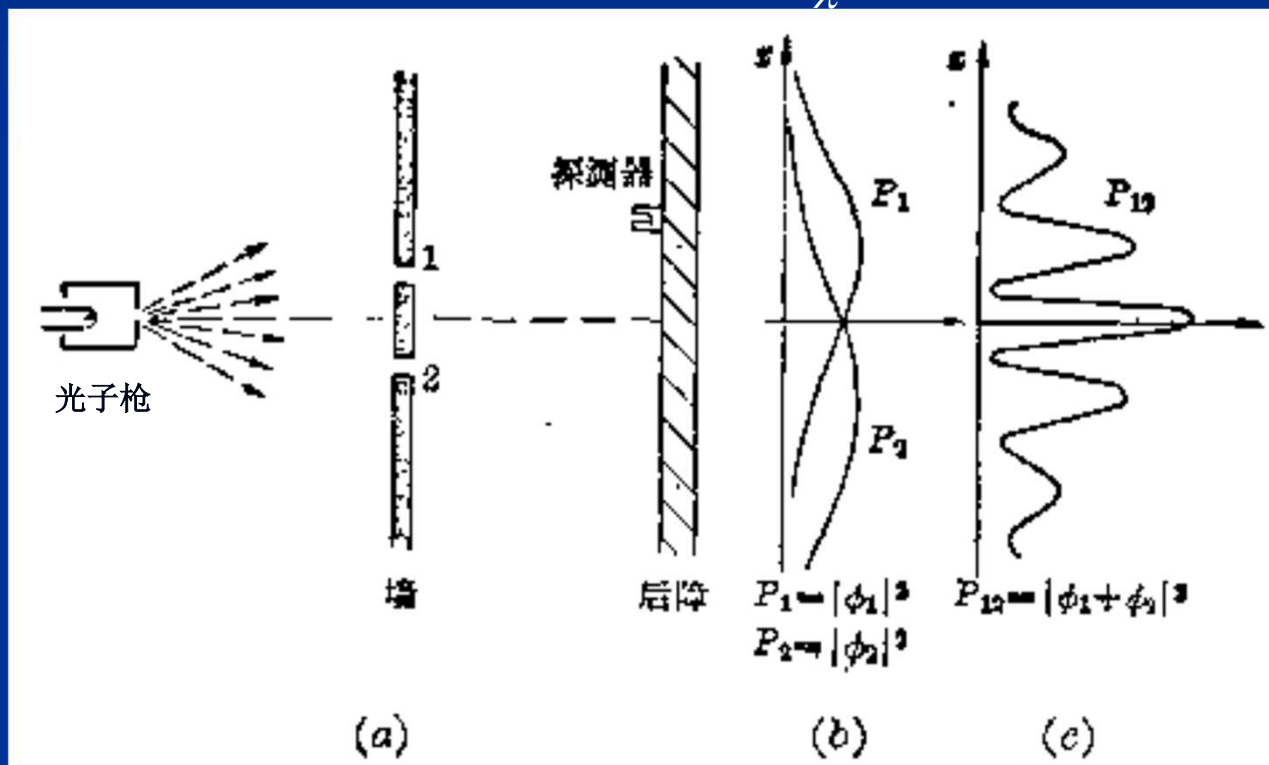
制作全息元件

9.7 光子波的干涉

光子：完整的能量颗粒

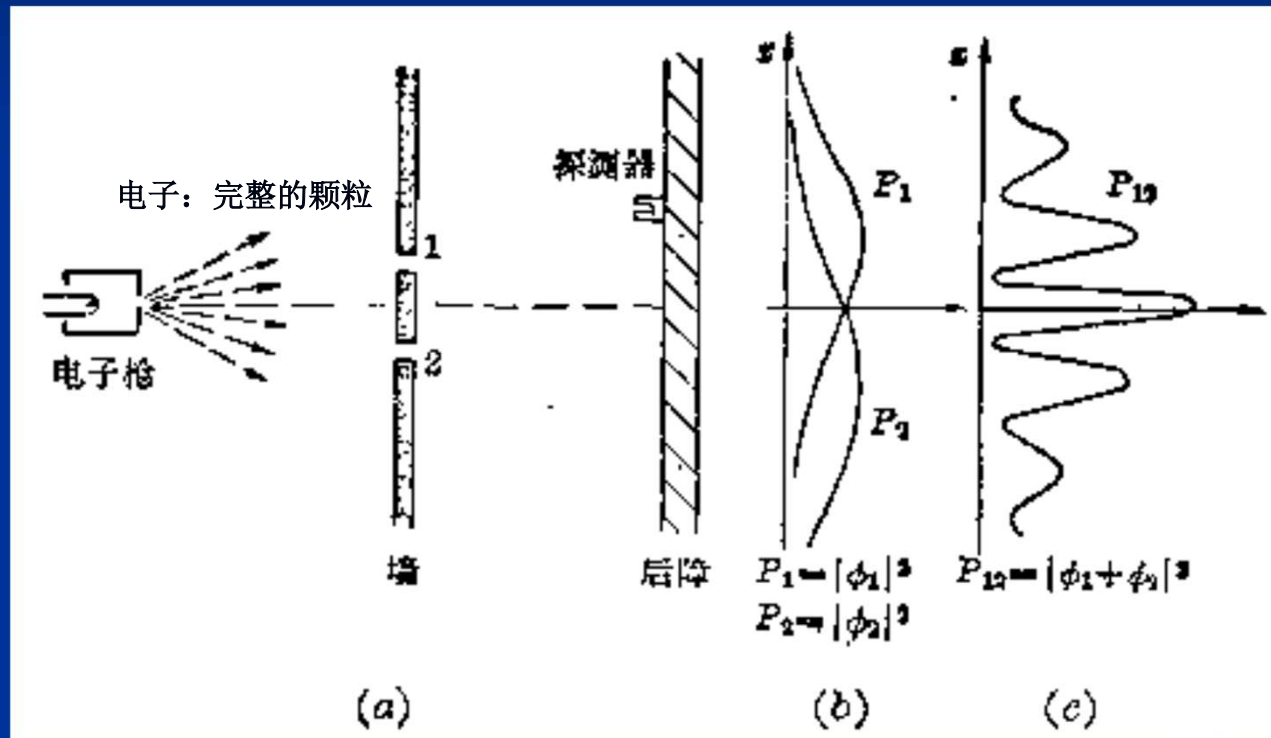
$$E = \hbar\omega = h\nu$$

$$\vec{p} = \hbar\vec{k} = \frac{h}{\lambda}\hat{k}$$



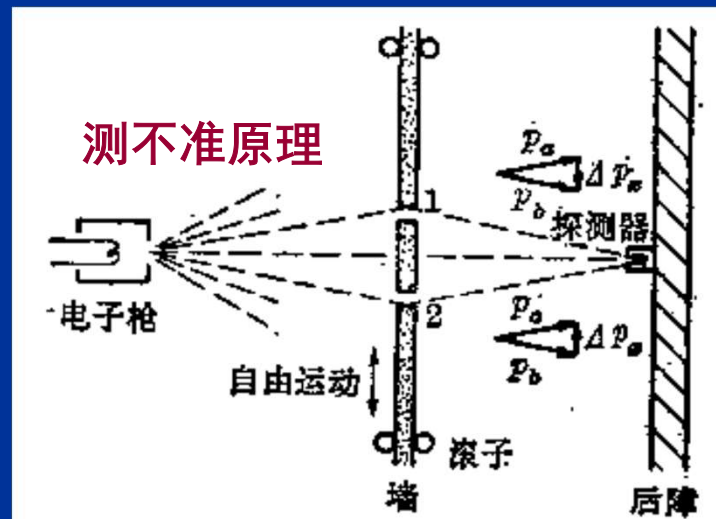
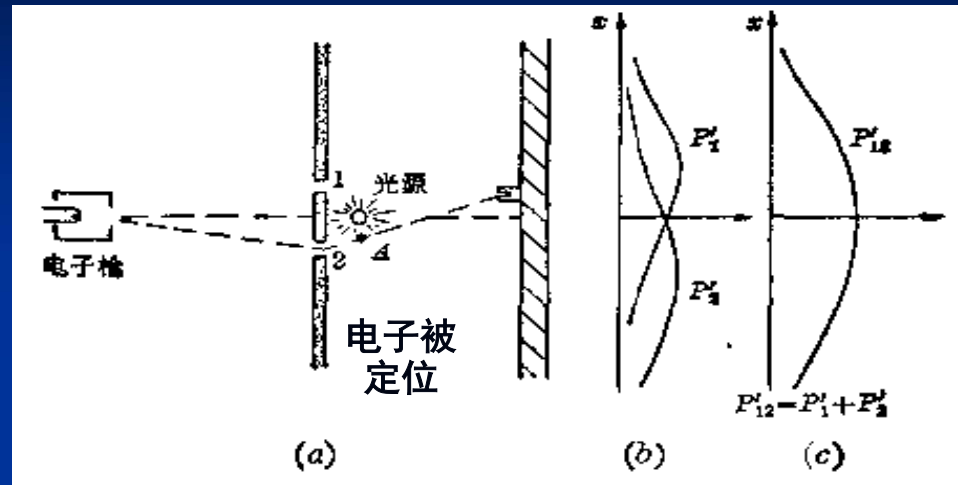
- * 某局域空间或某时刻：只能观察到一个个光子（迄今未发现半个光子）
- * 在大空间区域或在长时间范围内：波性分布（传统光波？）

9.8 电子波的干涉

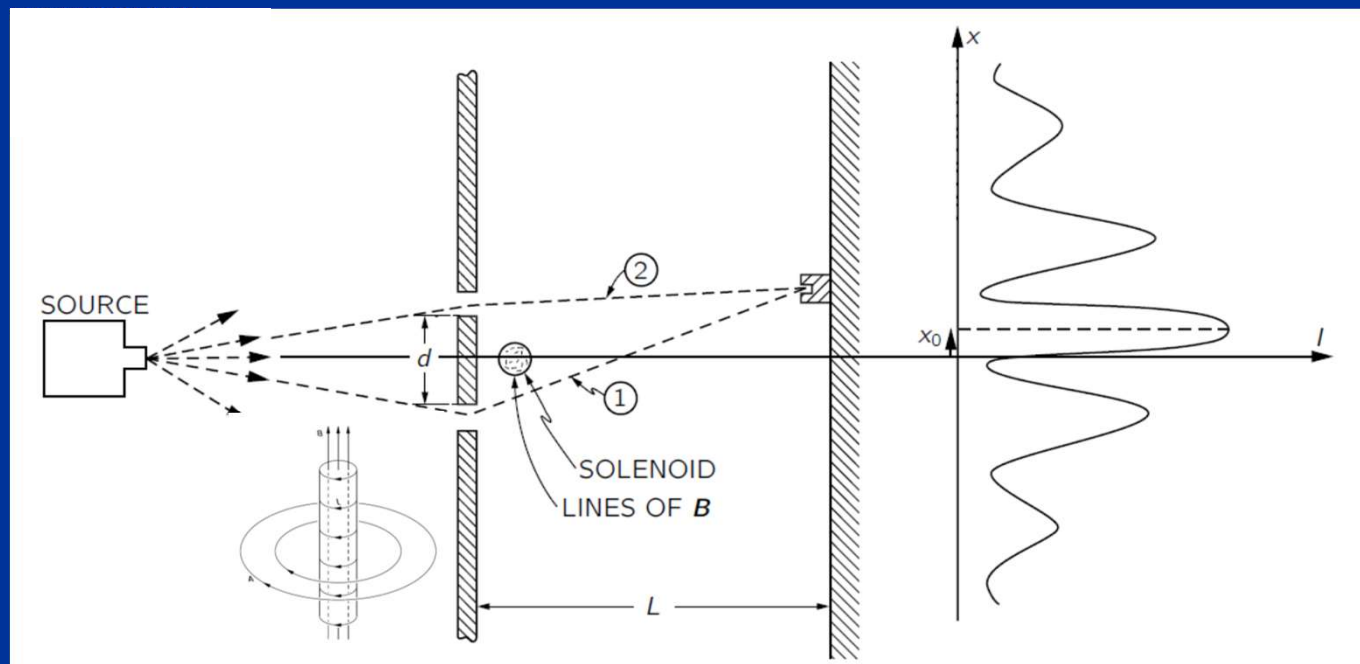
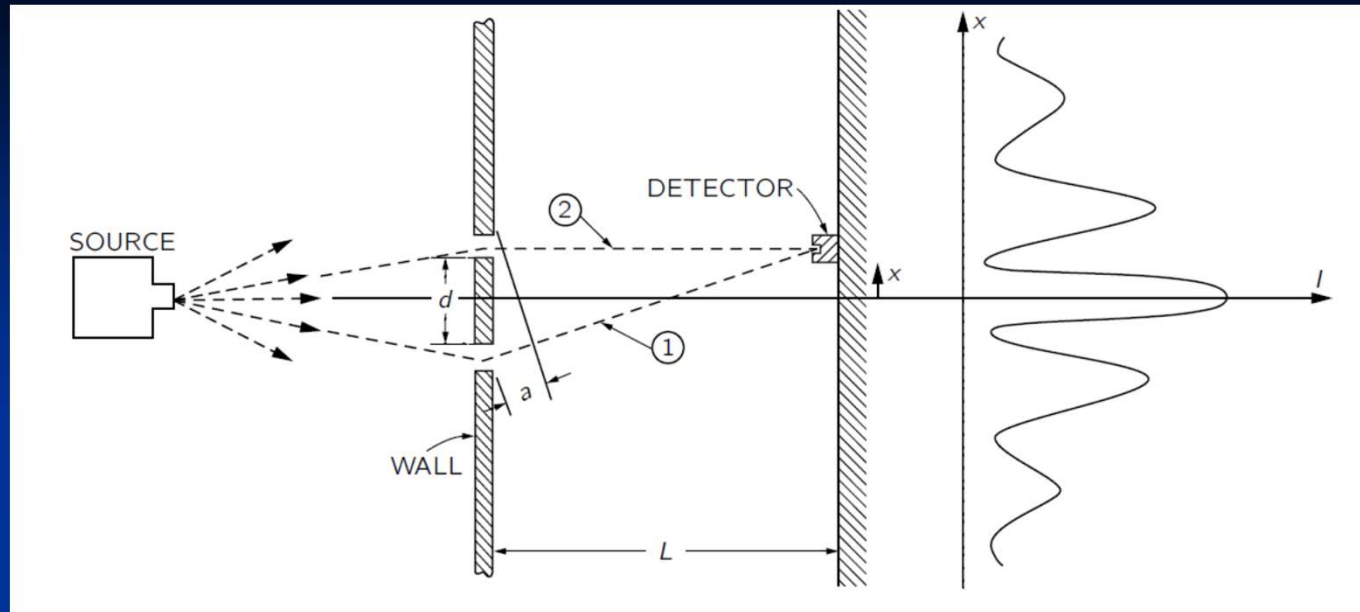


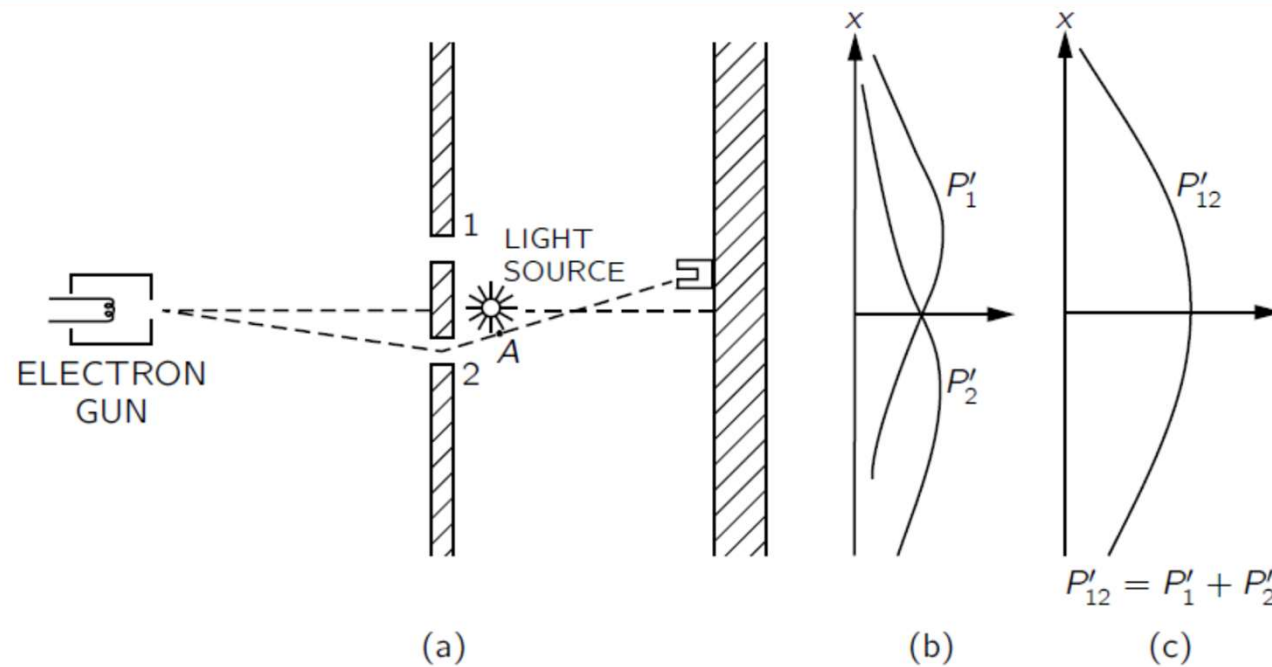
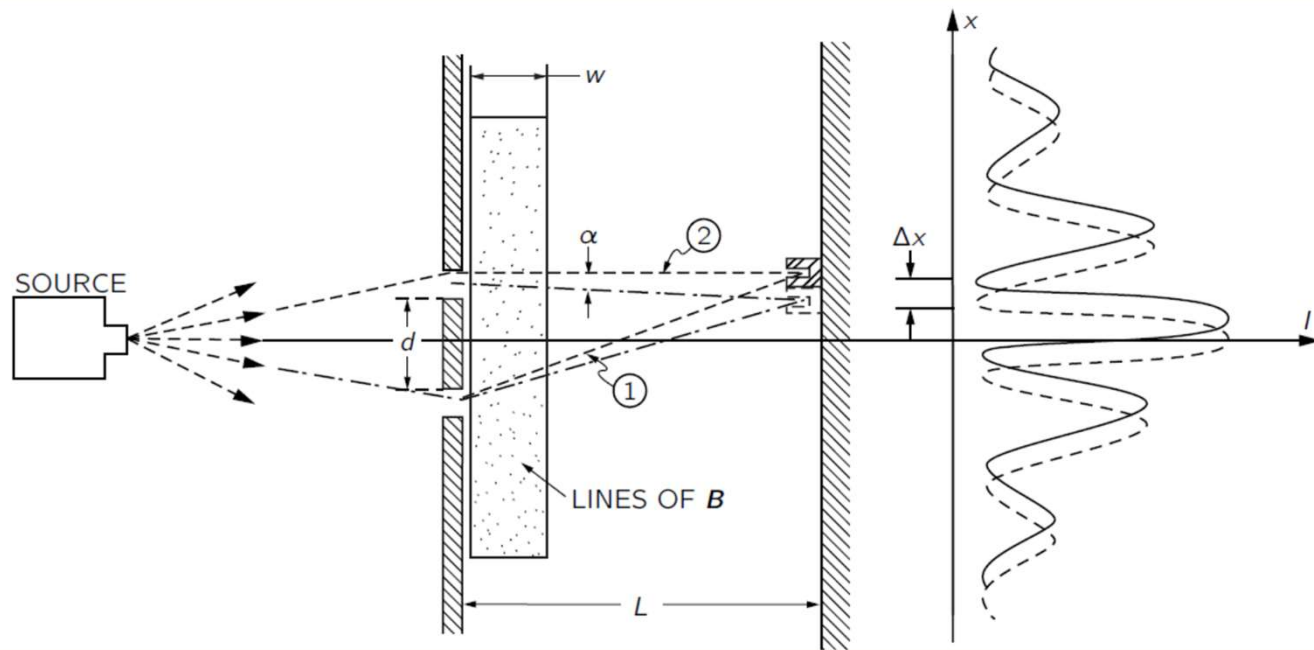
- * 某局域空间或某时刻: 只能观察到一个个电子 (迄今未发现半个电子)
- * 在大空间区域或在长时间范围内: 波性分布

“量子力学”被“测不准原理”严密保护



空间位置确定：动量成分大量增加，参与干涉的波函数大量增加，干涉图样弱化甚至消失
 动量确定：位置完全不确定，波函数的空间范围无限大





本章小结

解决干涉的：表征、特征、典型应用

- 干涉： 振幅加和后→平方 $\left(\sum_i \vec{E}_i(\vec{r}, t) \right)^2$
不干涉：振幅平方后→加和 $\left(\sum_i \vec{E}_i^2(\vec{r}, t) \right)$
- 干涉模型： 外差法、同频法
- 利用： 波前差、光程差
- 典型应用： 选谱、超衍射限精密测量